CHEMISTRY OF ADVANCED ENERGY STORAGE DEVICES FOR E-MOBILITY(18G6E14)

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Unit II: Lithium ion batteries

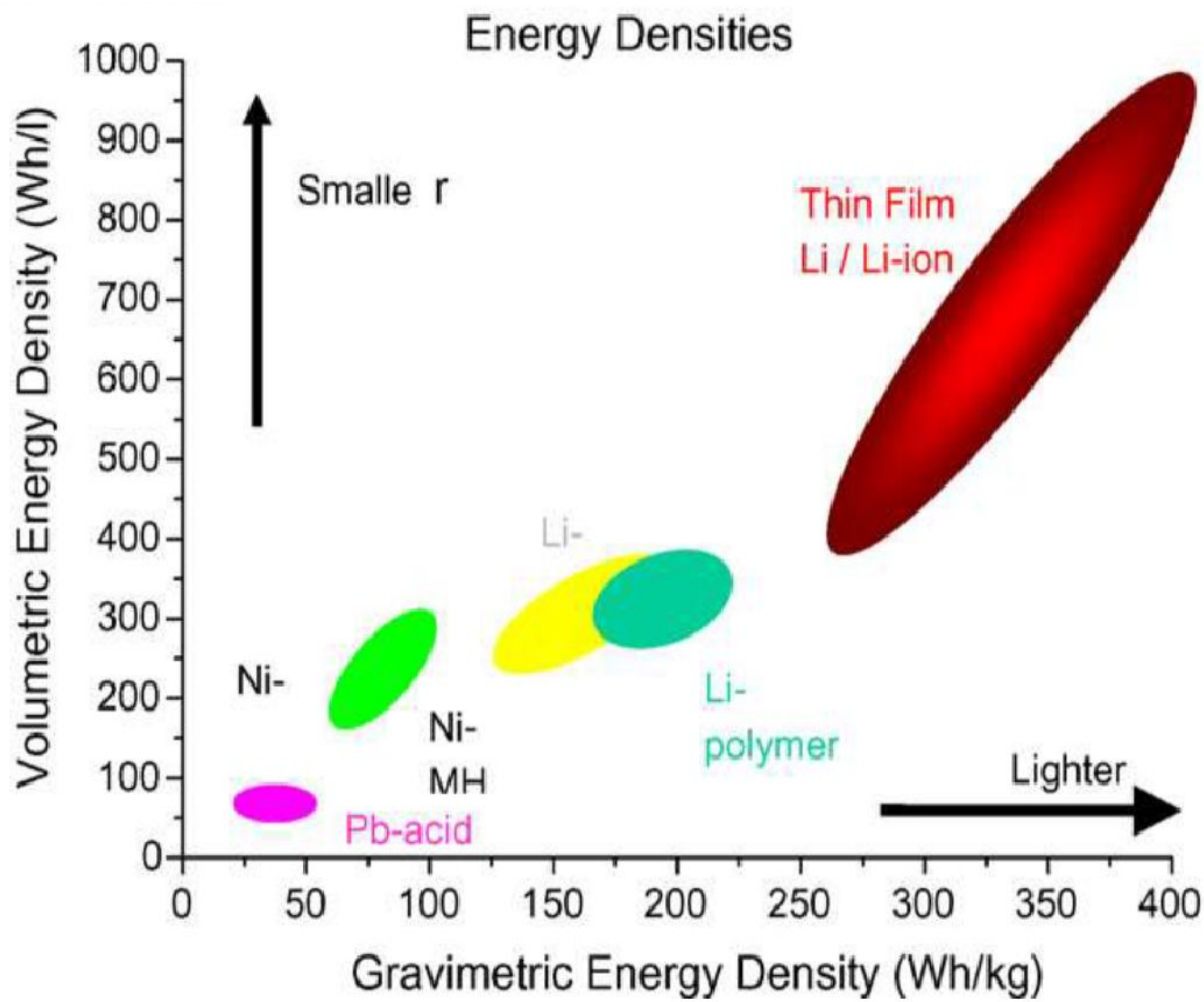
Criteria of battery systems for E mobility

- *High energy density*
- *High power density*
- *Good cycle life*
- *Wide temp. range of operation (-30 to +70^{o}C)*
- *Quick recharge*
- *Totally unassisted and maintenance free nature*
- *Tolerance to abuse*
- *Non-toxicity of battery materials*
- *Safety & reliability, non-polluting*

Why lithium for batteries?

- Lithium is the lightest of metal
- High capacity of storage of energy: 370 – 300 Wh/cm^3
- High electrochemical reduction potential
- Highly reactive material
- Spread thermal range -25˚C_+40˚C

Why lithium for batteries?

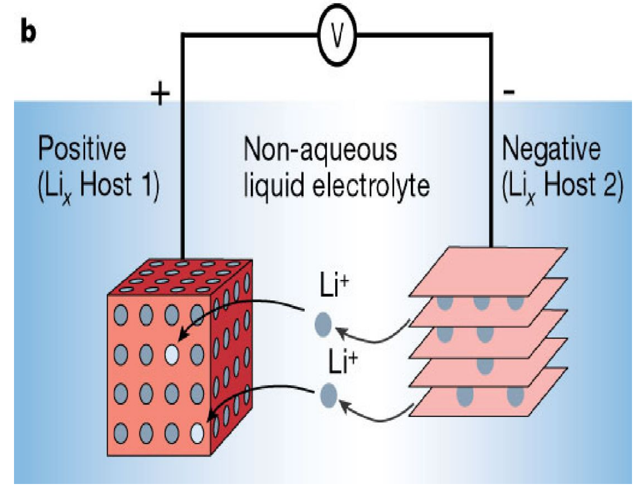


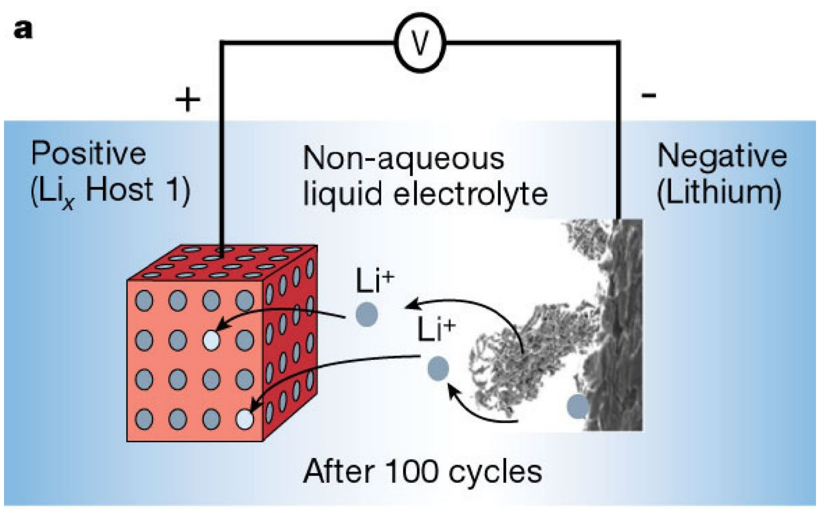
History of lithium batteries?

- The name of "lithium ion battery" was given by T. Nagaura and K. Tozawa
- The concept of "lithium ion battery" was firstly introduced by Asahi Kasei Co. Ltd
- Lithium ion batteries were first proposed by M. S. Whittingham in the 1970's. Whittingham used TiS_2 as the cathode and Lithium metal as the anode.

Contd…

Lithium Battery Development: Pioneering work for the lithium battery began in 1912 by G. N. Lewis but it was not until the early 1970's when the first non-rechargeable lithium batteries became commercially available. In the 1970's, Lithium metal was used but its instability rendered it unsafe.





Attempts to develop rechargeable lithium batteries followed in the eighties, but failed due to safety problems. The Lithium-Ion battery has a slightly lower energy density than Lithium metal, but is much safer. Introduced by Sony in 1991.

Types of lithium batteries?

- Primary batteries are disposable because their electrochemical reaction cannot be reversed. ΔG negative (irreversible): Eg $LiMnO_2$

- Secondary batteries are rechargeable, because their electrochemical reaction can be reversed by applying a certain voltage to the battery in the opposite direction of the discharge.

ΔG negative, discharge ΔG positive, charge

Eg: $LiCoO_2$, $LiFe(PO_4)_2$

Lithium ion secondary Batteries

- The lithium ion battery (LIB) system has been most successful in recent development of battery.
- Li is lightest metal and has one of the highest standard reduction potentials (-3.0 V)
- Theoretical specific capacity of 3860 Ah/kg in comparison with 820 Ah/kg for Zn and 260 Ah/kg for Pb.
- The first commercial lithium-ion battery was released by Sony in 1991
- Battery performance is related not only capacity but also to how fast current can be drawn from it: specific energy (Wh/Kg), energy density (Wh/cm^3) and power density (W/Kg)

Advantage and disadvantages of Lithium ion Batteries

Advantages of Lithium-ion batteries

POWER – High energy density means greater power in a smaller package.
160% greater than NiMH
220% greater than NiCd

HIGHER VOLTAGE – a strong current allows it to power complex mechanical devices.

LONG SHELF-LIFE – only 5% discharge
loss per month.
10% for NiMH, 20% for NiCd

Disadvantages of Lithium-ion batteries

EXPENSIVE -- 40% more than NiCd

DELICATE -- battery temperature must be monitored from within (which raises the price), and sealed particularly well

REGULATIONS -- when shipping Lithium-ion batteries in bulk (which also raises the price)
Class 9 miscellaneous hazardous material

Anode materials

Desired characteristics of anode materials

1) Large capability of Lithium adsorption

2) High efficiency of charge/discharge

3) Excellent cyclability

4) Low reactivity against electrolyte

5) Fast reaction rate

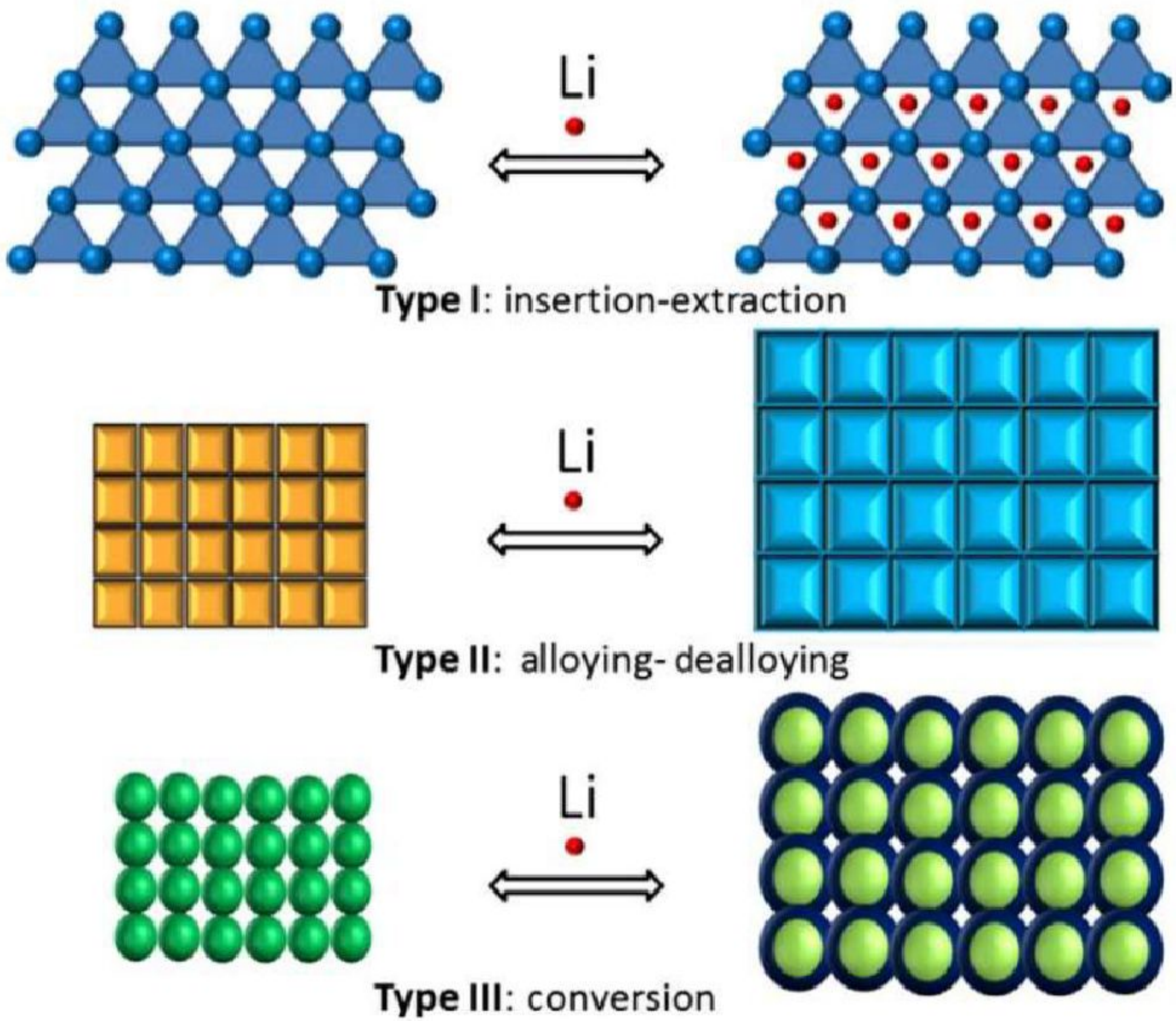
6) Low cost

8) Environmental -friendly, non-toxic

- Commercial anode materials:

Hard Carbon, Graphite

Anode materials for LIB



Insertion type anode materials

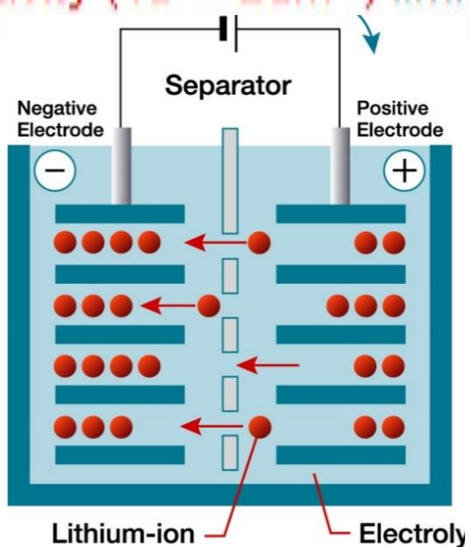
Titanium Based Oxides

$Li_4Ti_5O_{12}$ is considered the most appropriate titanium based oxide material for lithium storage purposes because

1. It exhibits excellent Li-ion reversibility at the high operating potential of 1.55 V vs. Li/Li+.
2. Lithium insertion/extraction in LTO occurs by the lithiation of spinel $Li_4Ti_5O_{12}$ yielding rock salt type $Li_7Ti_5O_{12}$.
3. During the insertion process, the spinel symmetry and its structure remain almost unaltered.

Limitation:

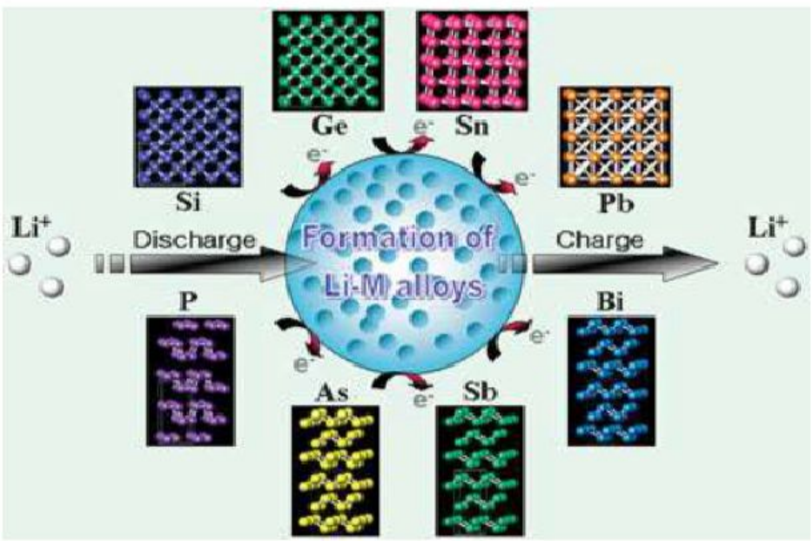
Its poor electronic conductivity (10^{-13} Scm^{-1}) limits its full capacity at high charge and discharge.

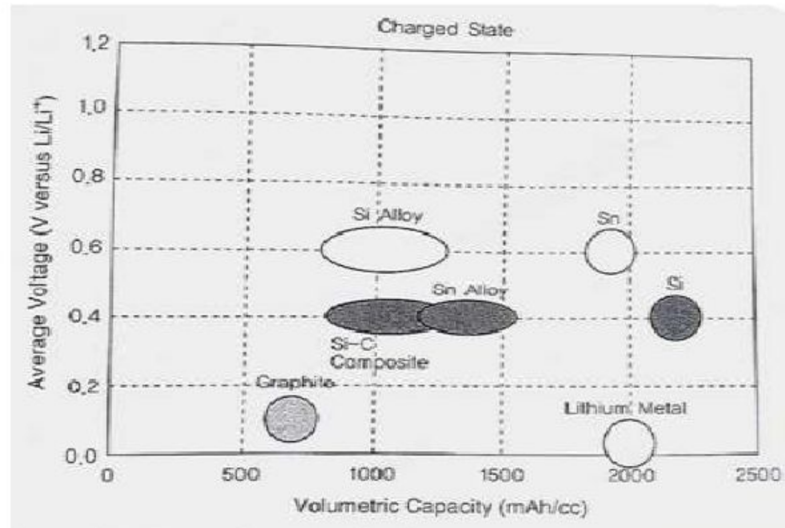


Alloy/dealloy type anode materials

Which can react with lithium to form alloys

$xLi^+ + xe^- + M \leftrightarrow Li_xM$





Voltage and capacity of metal elements that form alloys with lithium.

The working voltage of Sn and Sn alloys at 0.6V vs lithium is 0.2V higher than that of Si and Si alloys

Tin based anode materials

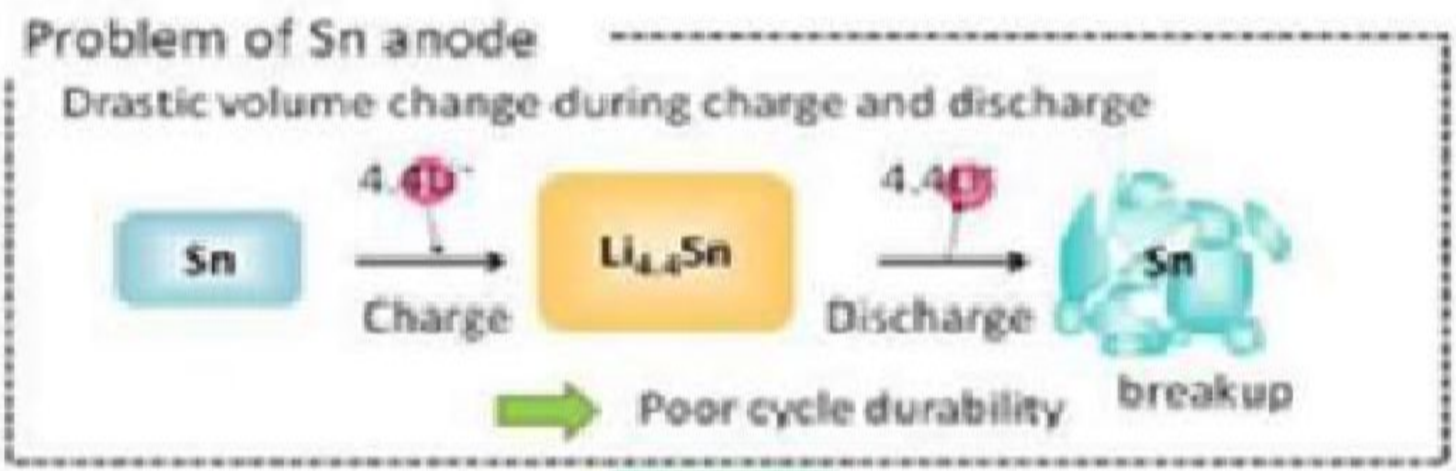
Tin oxide (SnO_2):

Reversible Sn-lithium alloying/ dealloying reaction:

$SnO_2 + 4Li \leftrightarrow Sn + 2Li_2O$,

$Sn + 4.4Li^+ \leftrightarrow Li4.4Sn$

This overall electrochemical process involves 8.4Li for one SnO_2 formula unit.



Cathode Materials

Desired characteristics of Cathode materials

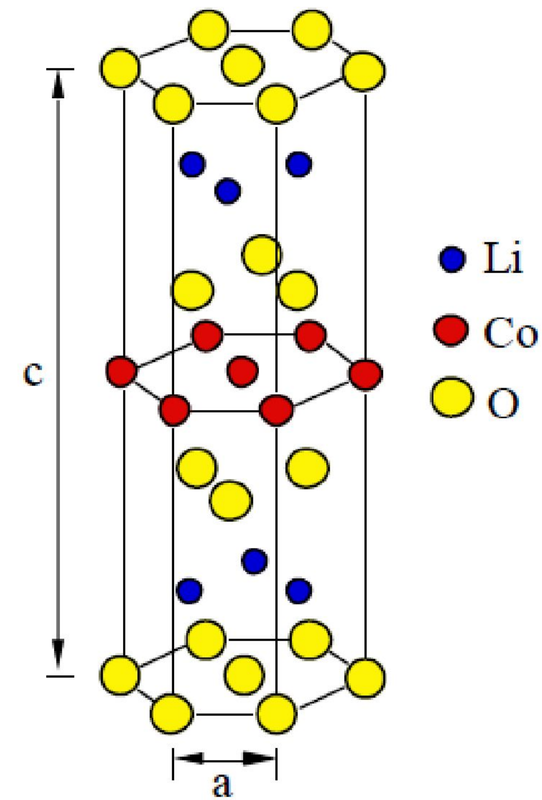
- High discharge voltage
- High energy capacity
- Long cycle life
- High power density
- Light weight
- Low self-discharge
- Absence of environmentally hazardous elements

- Commercial cathode materials:

$LiCoO_2$, $LiMn_2O_4$, $LiNiO_2$, $LiFePO_4$

Parameters effecting Cathode behaviour

- Method of preparation
- Particle size
- Morphology
- Oxygen Deficiency
- Temperature



$LiCoO_2$

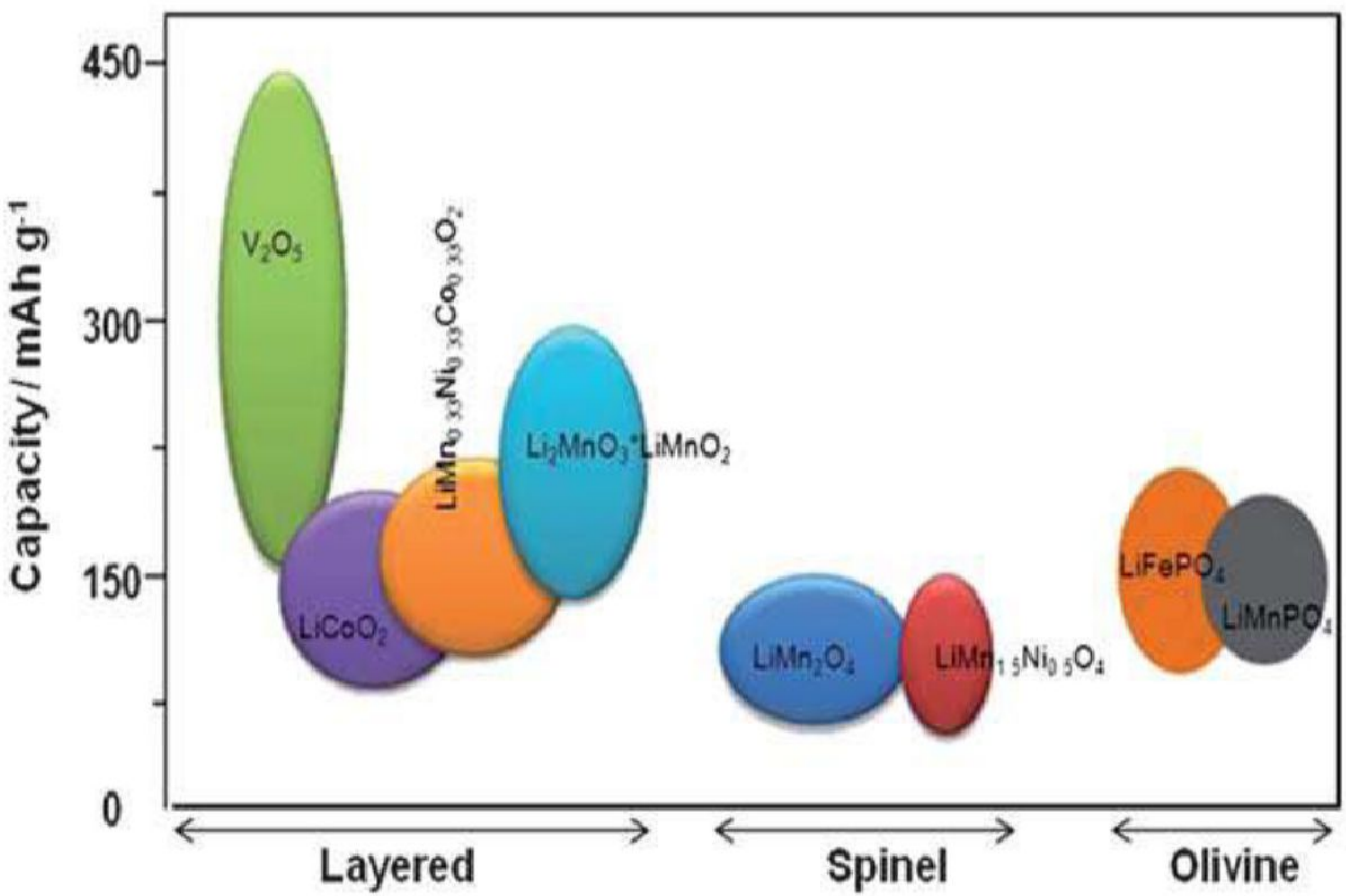
Advanced materials used as cathode Layered oxide cathodes

- Spinel oxide cathodes
- Zigzag layered $LiMnO_2$ compound
- Olivine structure of $LiMPO_4$
- Other compounds

Important cathode materials in research

Material	Potential (V)	Specific capacity (Ah/kg)	Energy (Wh/kg)
$LiCoO_2$	3.9	130	507[1]
$LiMn_2O_4$	3.95	148	585[1]
FeF_3	2.74	712	1951[2]
BiF_3	3.13	302	945[2]
MnF_3	2.65	719	1905[2]
CuF_2	3.55	528	1874[2]
$LiFePO_4$	3.4	170	578[3]
$LiMnPO_4$	4.1	171	701[3]
$LiCoPO_4$	4.8	167	802[3]
$LiNiPO_4$	5.1	167	852[3]
Li_2FeSiO_4	3.3	328	1082
Li_2MnSiO_4	4.0	333	1332
Li_2CoSiO_4	4.3	325	1397
Li_2NiSiO_4	4.7	325.5	1530

Comparison of capacity of different cathode materials



Electrolytes

- Role of electrolyte

1) Ion conductor between cathode and anode
2) Generally, Lithium salt dissolved in organic solvent
3) Solid electrolyte is also possible if the ion conductivity is high at operating temperature.

- **Characteristics of Electrolyte**

1) Inert
2) High ionic conductivity, low viscosity
3) low melting point & high dielectric constant (ε)
4) Appropriate concentration of Lithium salt
5) Chemical/thermal stability, High flash point (Tf), nontoxic,
6) Low cost
7) Environmental -friendly, non-toxic

- Commercial electrolytes: $LiPF_6$ in Carbonate solvent, propylene carbonate, 1,2 dimethoxy ethane

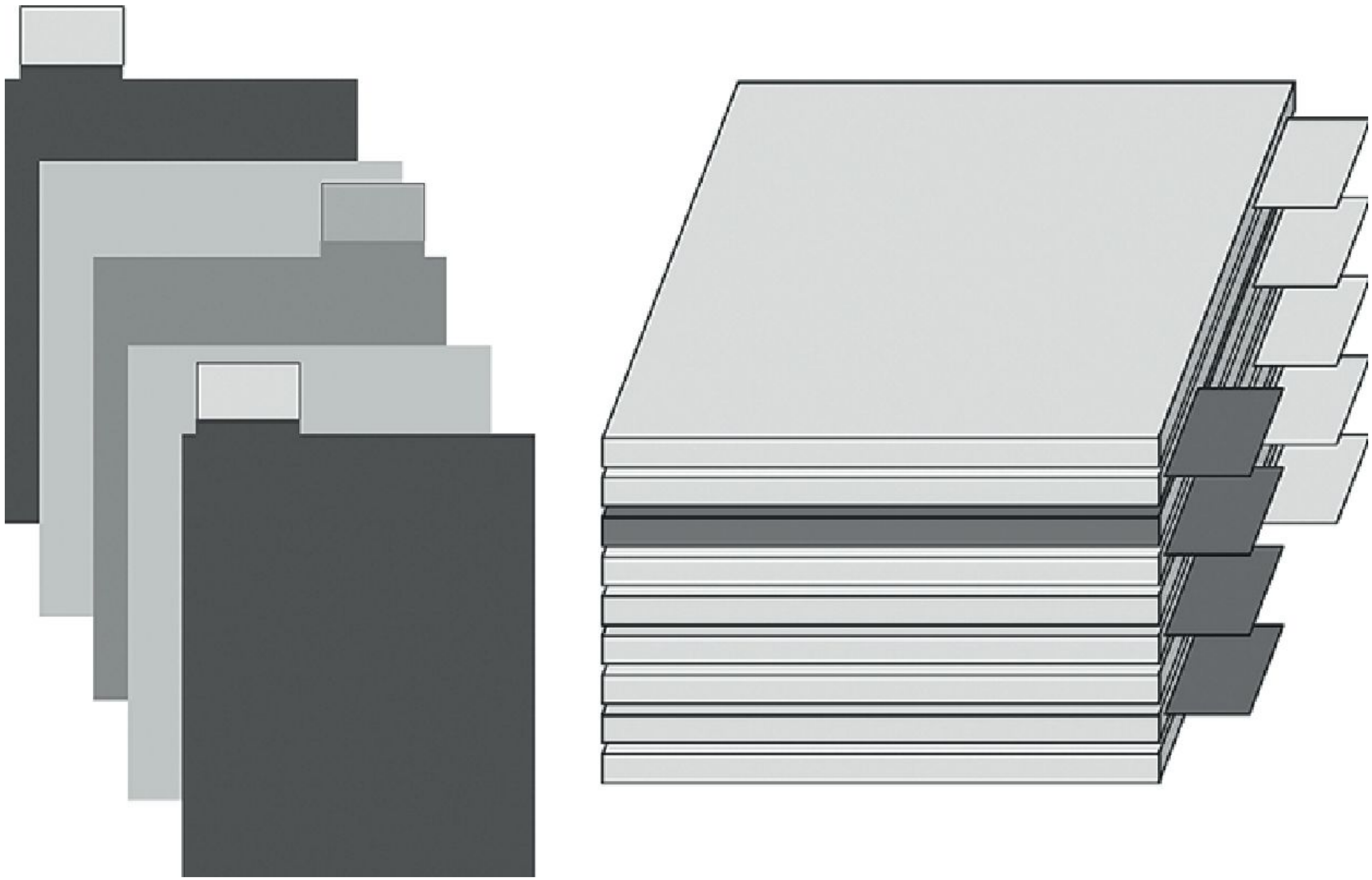
Types of Electrolytes

1) Liquid electrolyte: $LiPF_6$ in 1,2, dimethoxy ethane
2) Molten lithium salt: LiCl, LiBr etc
3) Amorphous Polymer Electrolytes: Lithium salts in Propylene carbonate
4) Crystalline Polymers:

Electrolyte additives

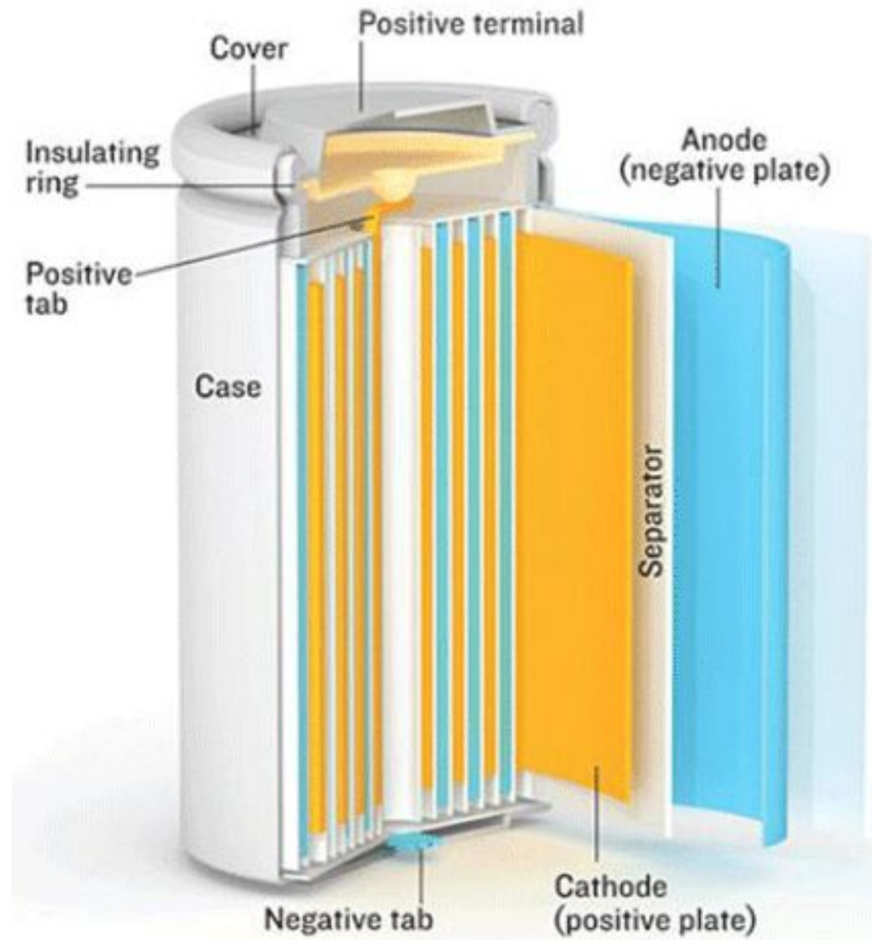
1) Those used for improving the ion conduction properties in the bulk electrolytes
2) Those used for SEI chemistry modifications
3) Those used for preventing overcharging of the cells

Fabrication of battery electrodes



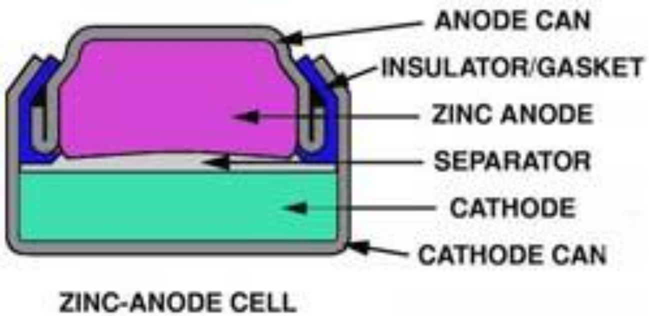
Cylindrical cell

- Classic packaging for primary & secondary cells
- High mechanical stability, economical, long life
- Holds internal pressure without deforming case
- Inefficient use of space
- Metal housing adds to weight



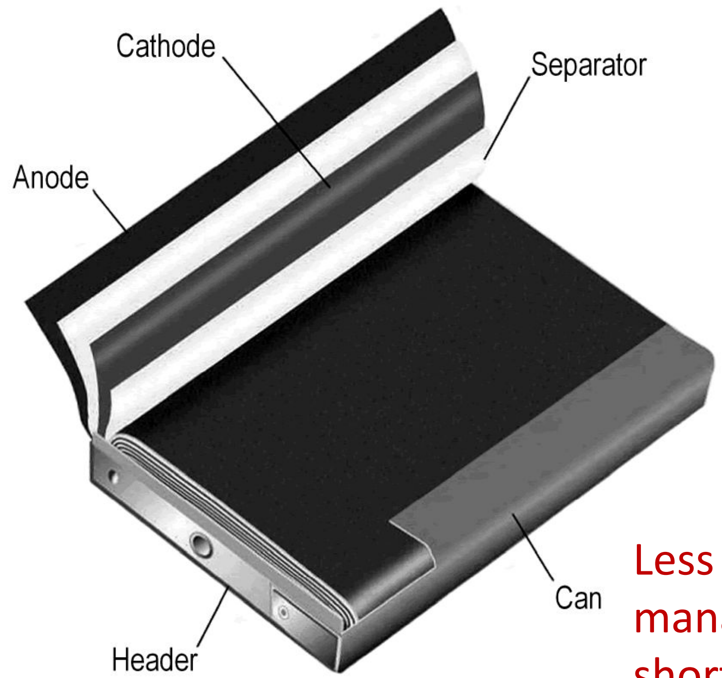
Button cell

- Also known as coin cells; small size, easy to stack
- Mainly reserved as primary batteries in watches, gauges
- Rechargeable button cells do not allow fast charging
- Limited new developments
- Must be kept away from children, harmful if swallowed (voltage)

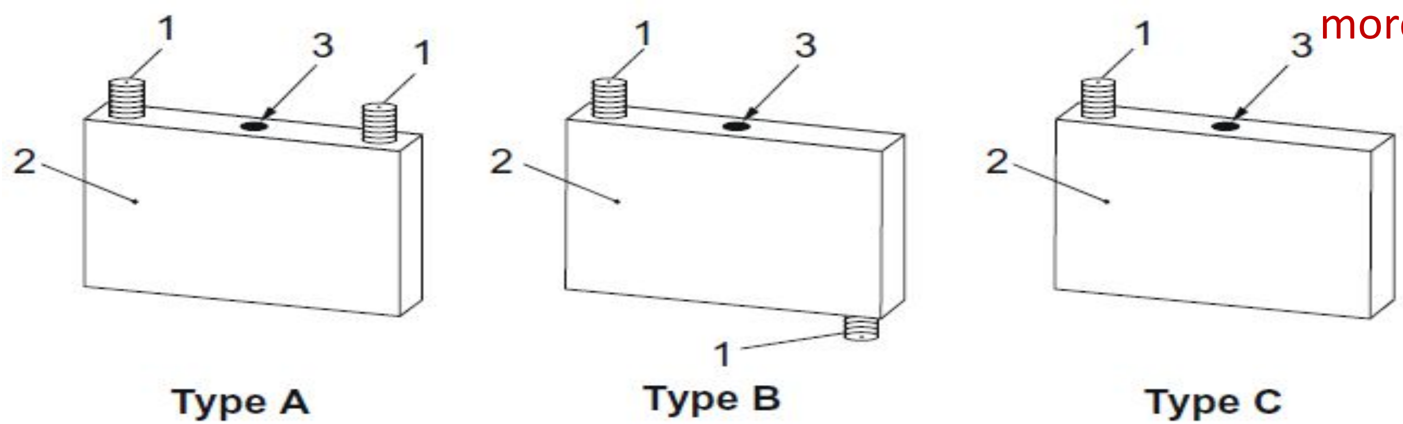


Prismatic cell

- Best usage of space
- Allows flexible design
- Higher manufacturing cost
- Less efficient thermal management
- Shorter life

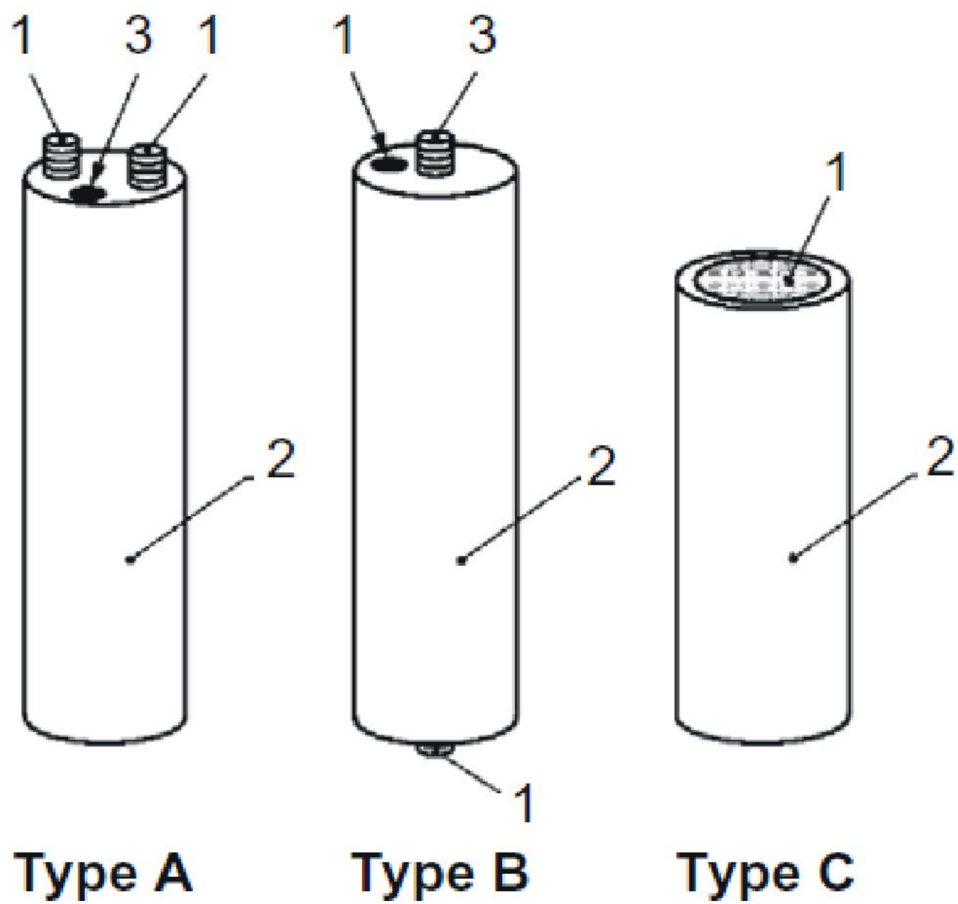


Less efficient in thermal management; possible shorter cycle life; can be more expensive to make.



Prismatic cells (1) Terminal, (2) cell housing, and (3) overpressure safety device.

Cell fabrication/modules

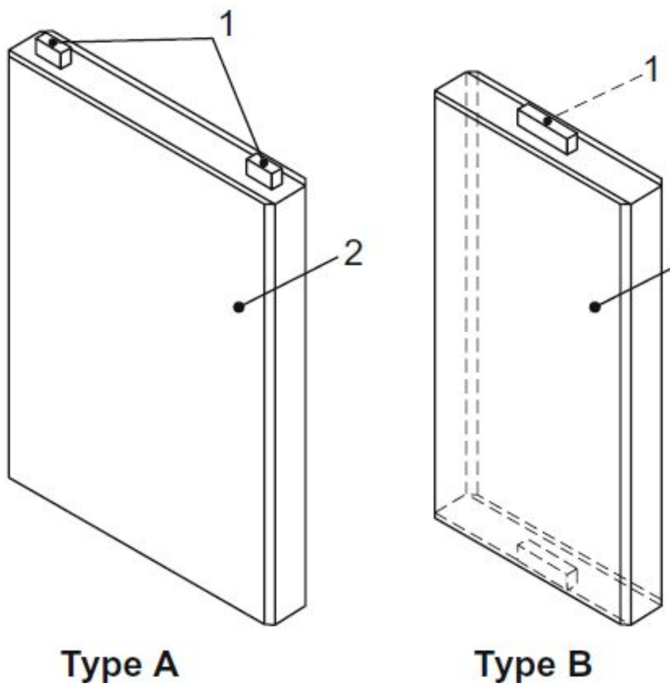


Cylindrical cells 1) Terminal, (2) cell housing, and (3) overpressure safety device.

Heavy; creates air gaps on multi-cell packs. Not suitable for slim designs.

Standard sizes: 18650, D, AA…
Steel casing
Low manufacturing cost
High specific energy (Wh/kg)
Good mechanical stability

Cell fabrication/modules and packs



Pouch cells (1) Terminal and (2) cell housing.

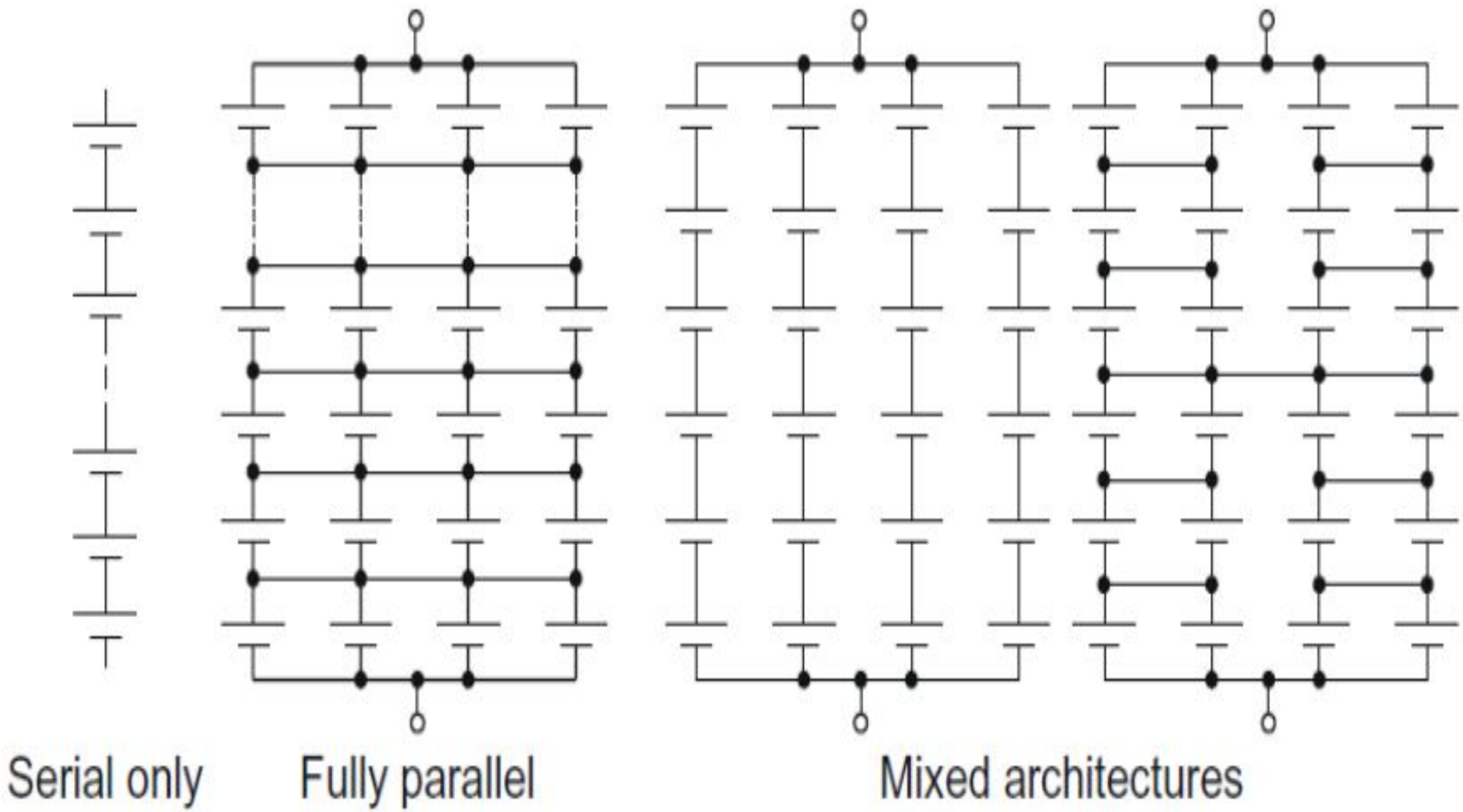
No standard size, each manufacturer designs its own
Laminated bag
High energy density (Wh/L)
Requires stacking pressure
Sensitive to moisture and high pressure

Exposure to humidity and heat shorten service life; 8–10% swelling over 500 cycles.

General battery packs components

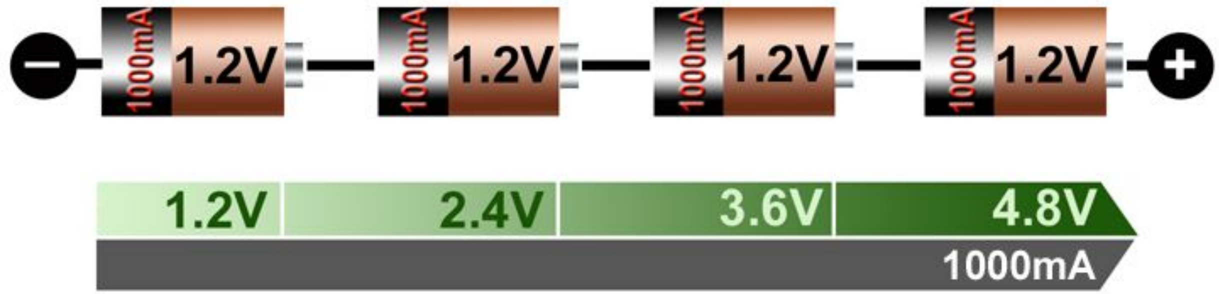
- **Battery system** designed for (P)HEV, EV
  - **Cells** designed for (P)HEV, EV
  - **Electric wiring harness, fuse, switch**
  - **Battery management system**
  - **Mechanical components, housing**
  - **Cooling, heating**

Designing of battery packs



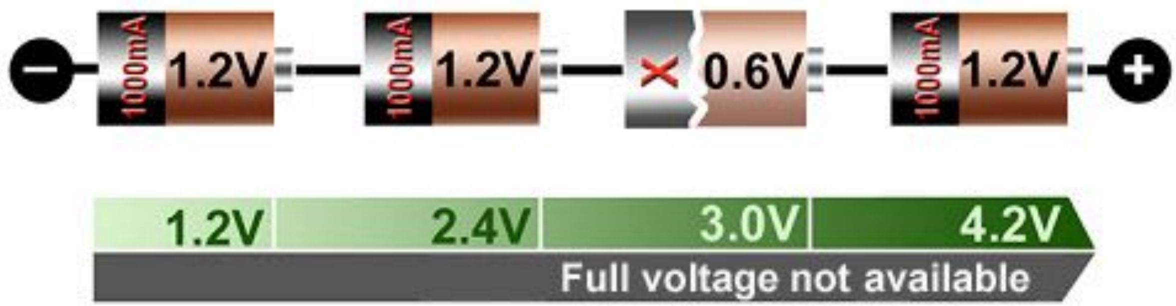
Series connection

Good string



- Adding cells in a string increases voltage; same current

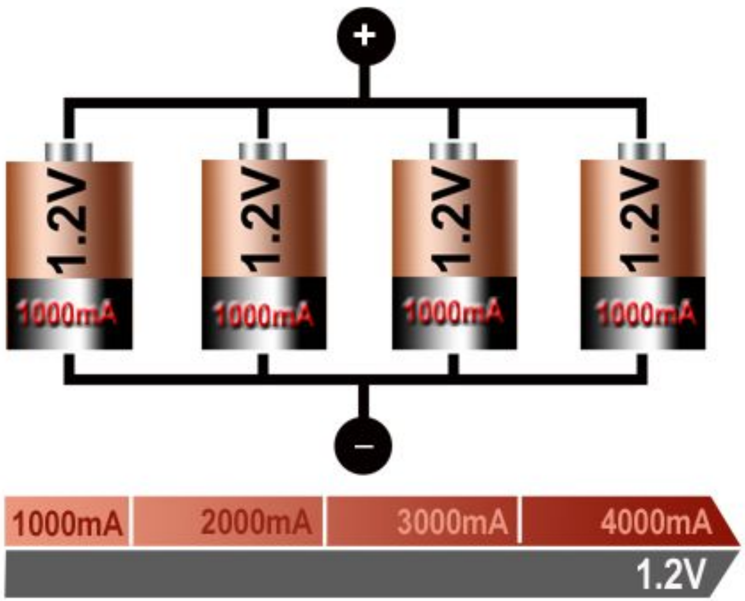
Faulty string



- Faulty cell lowers overall voltage, causing early cut-off
- Weakest cell is stressed most; stack deteriorates quickly

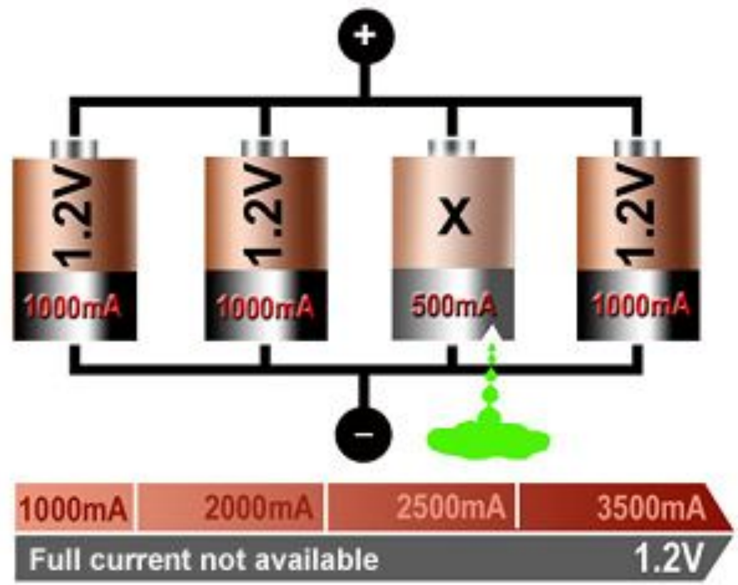
Parallel connection

Good parallel pack



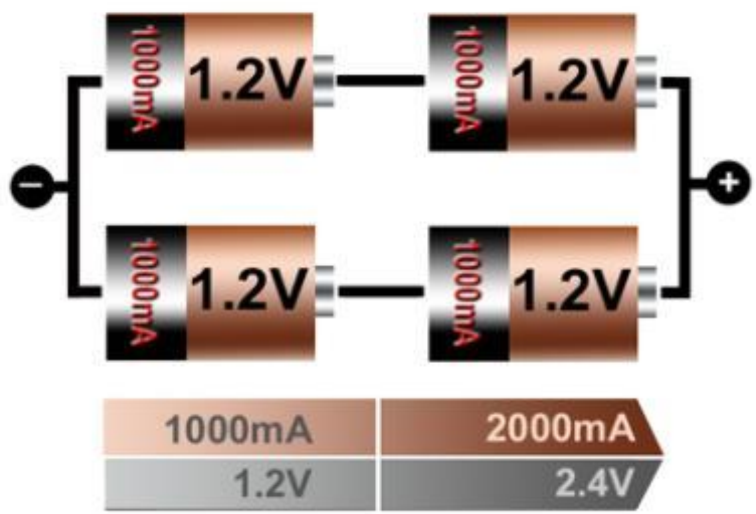
Allows high current;
same voltage

Faulty parallel pack



Weak cell reduces current,
poses a hazard if shorted

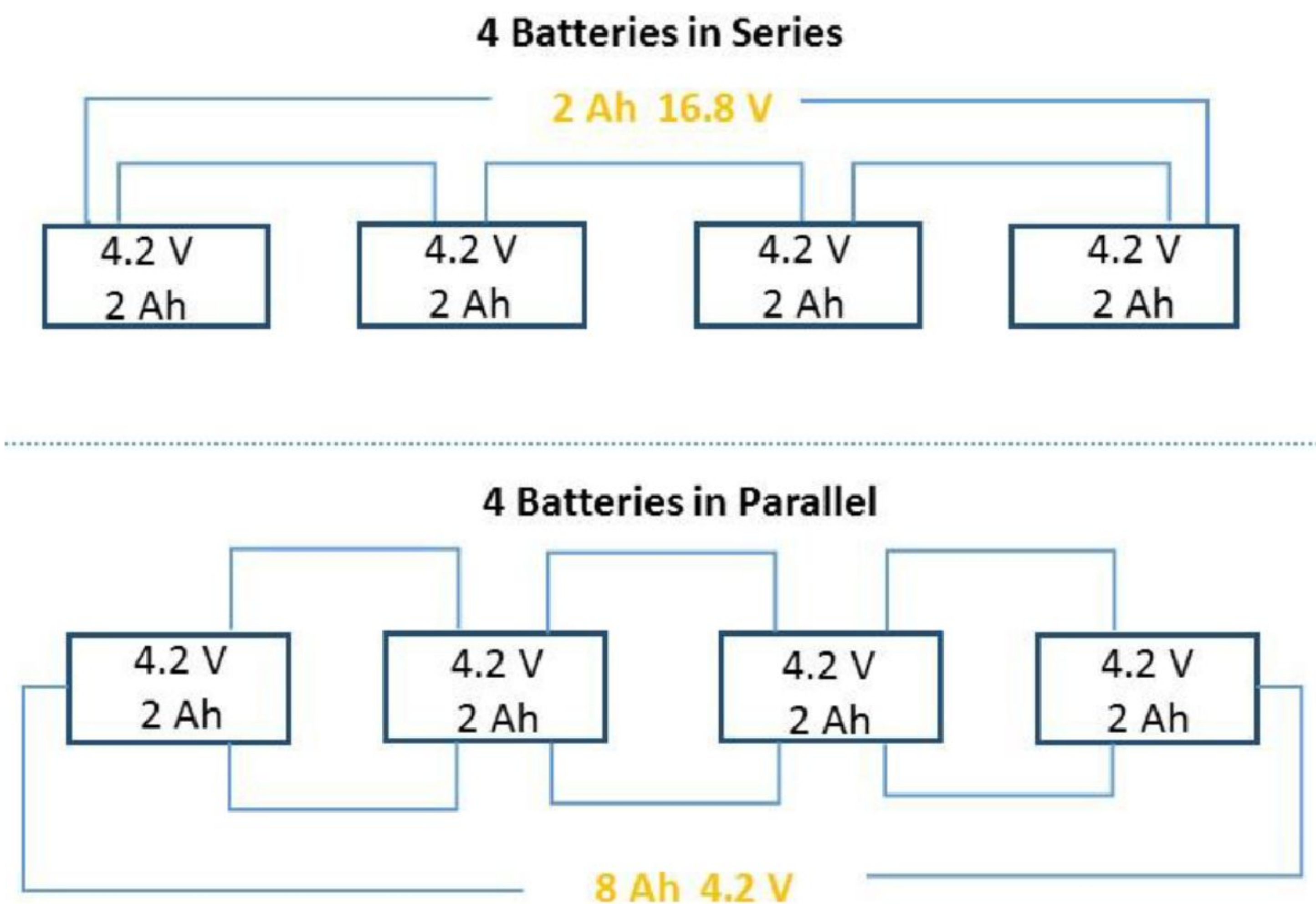
Serial- Parallel connection



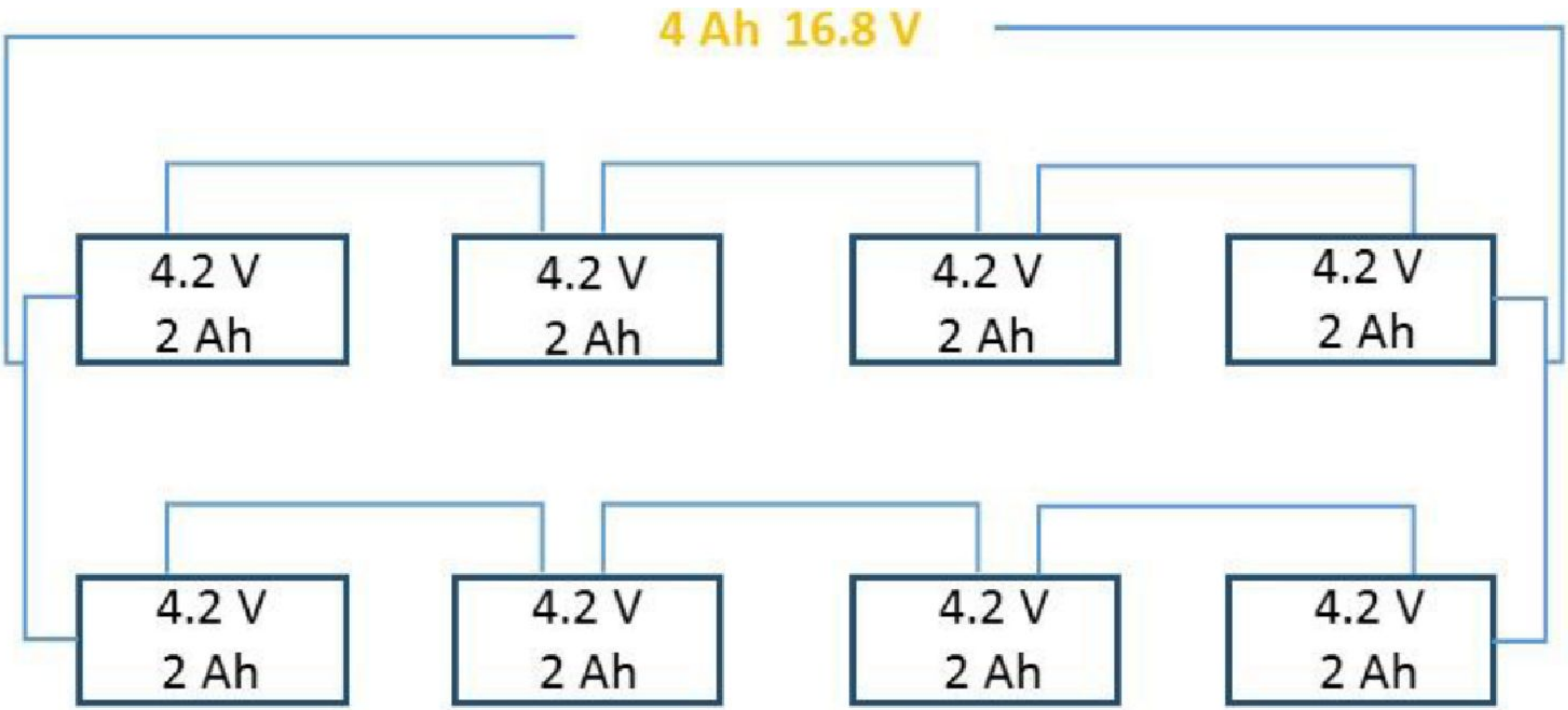
2S2P means:
2 cells in series
2 cells in parallel

- Most battery packs have serial-parallel configurations
- Cells must be matched

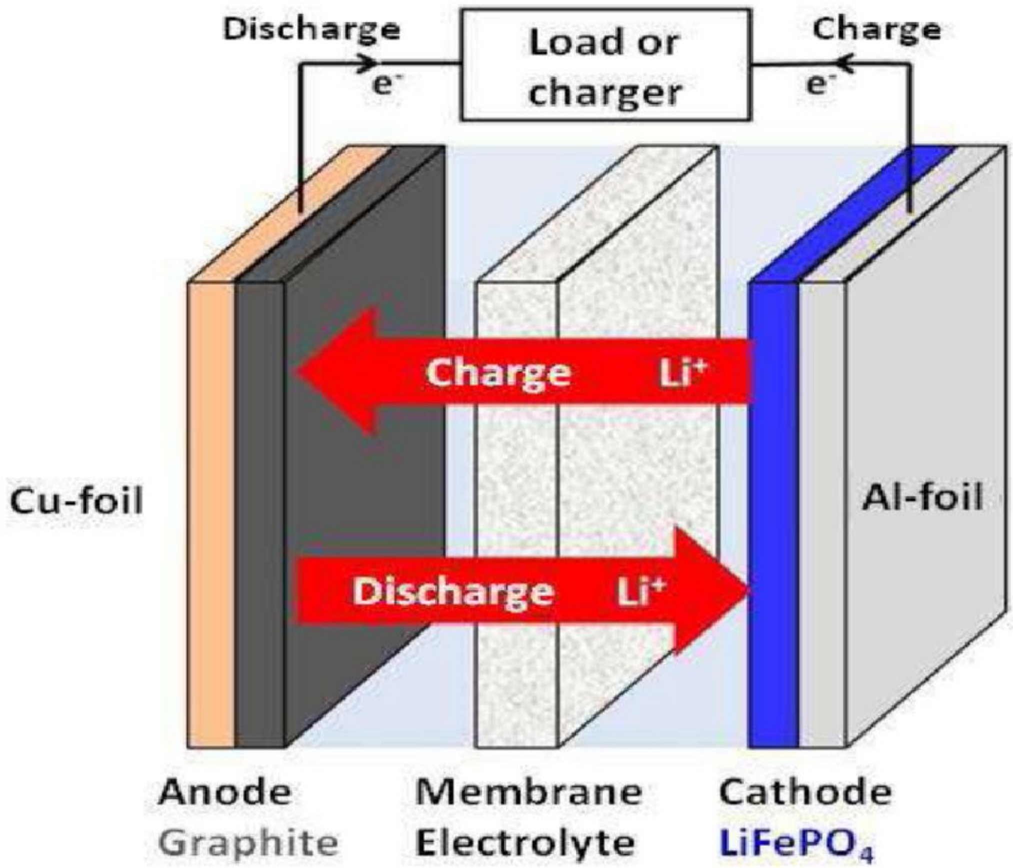
Designing of battery connections



Designing of battery connections

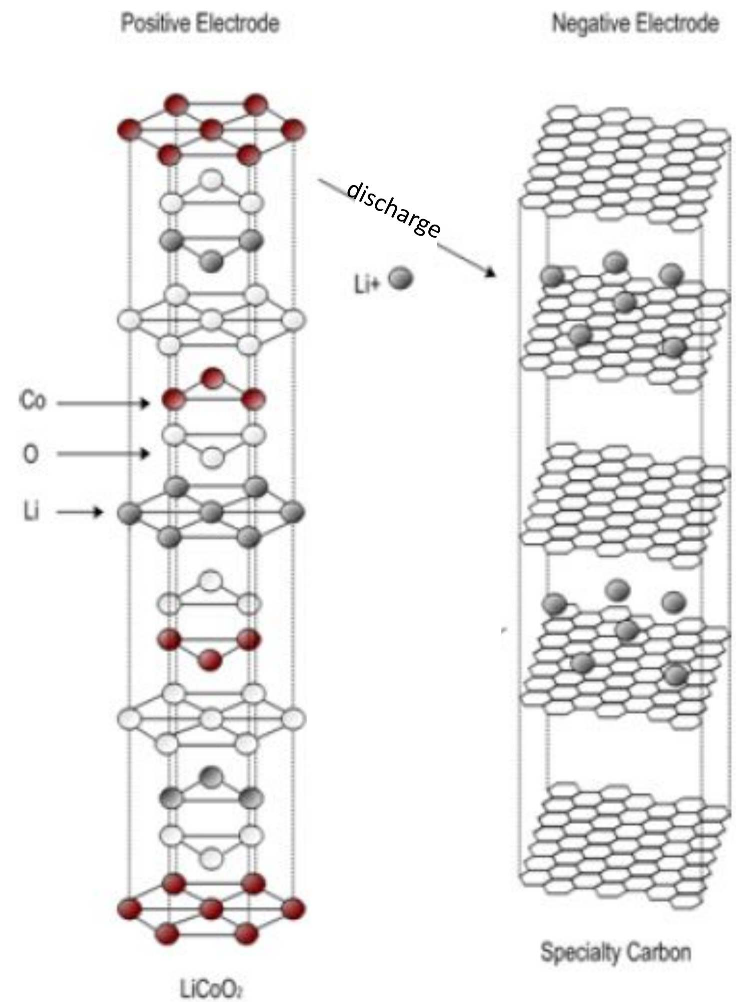


Construction and working of Li-ion batteries



Construction and working of Li-ion batteries

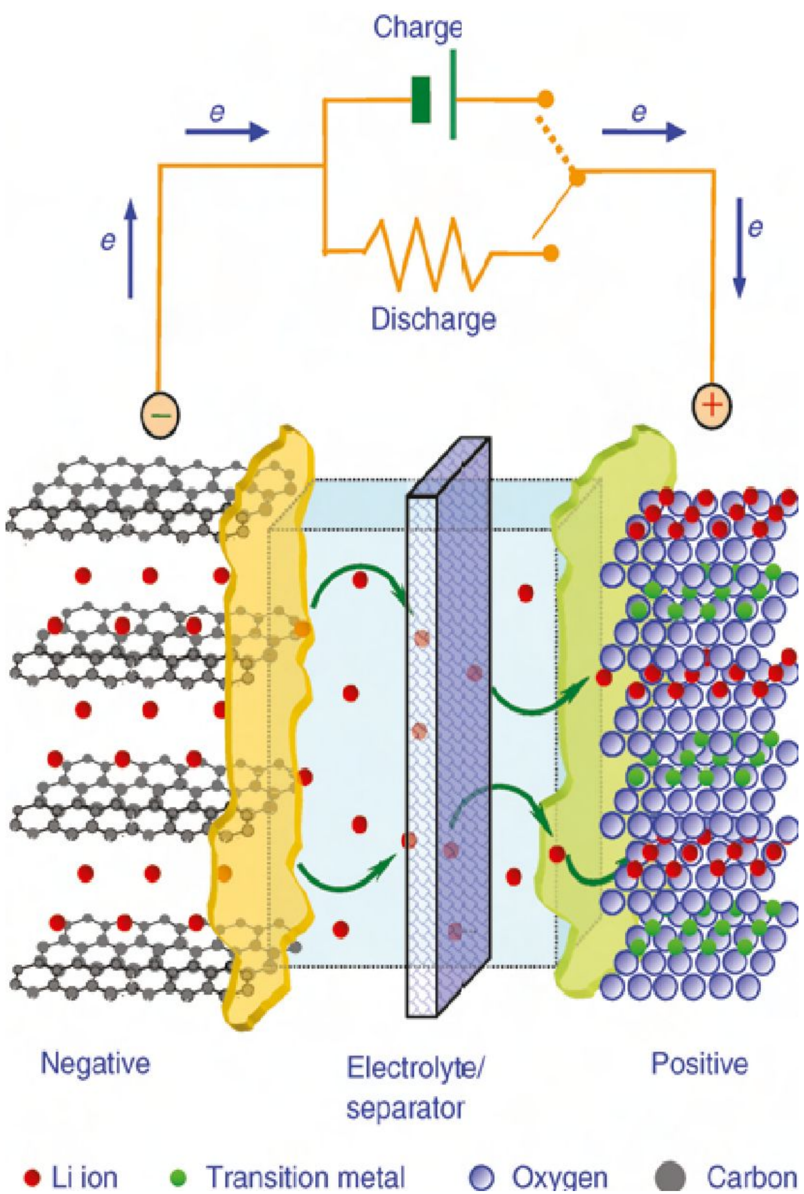
•Positive electrode: Lithiated form of a transition metal oxide (lithium cobalt oxide-$LiCoO_2$ or lithium manganese oxide $LiMn_2O_4$)



• Negative electrode: Carbon (C), usually graphite (C_6)

• Electrolyte: solid lithium-salt electrolytes ($LiPF_6$, $LiBF_4$, or $LiClO_4$) and organic solvents (ether)

Li-ion battery system



Electrochemical Reactions

- Cathode

$$LiCoO_2 \underset{d}{\overset{c}{\rightleftharpoons}} Li_{1-x}CoO_2 + xLi^+ + x\,e^-$$

- Anode

$$C_n + xLi^+ + x\,e^- \underset{d}{\overset{c}{\rightleftharpoons}} C_nLi_x$$

- Overall

$$LiCoO_2 + C_n \underset{d}{\overset{c}{\rightleftharpoons}} Li_{1-x}CoO_2 + C_nLi_x$$

Cell reaction/Working of Li-ion batteries

- Chemical reaction (charging)

  - Positive electrode $$LiCoO_2 \longrightarrow Li_{1-x}CoO_2 + xLi^+ + xe^-$$

    Through electrolyte

    Through load

  - Negative electrode $$xLi^+ + xe^- + 6C \longrightarrow Li_xC_6$$

  - Overall $$LiCoO_2 + C_6 \longrightarrow Li_{1-x}CoO_2 + C_6L_x$$

- In the above reaction *x* can be 1 or 0

- With discharge the Co is oxidized from Co^{3+} to Co^{4+}. The reverse process (reduction) occurs when the battery is being charged.

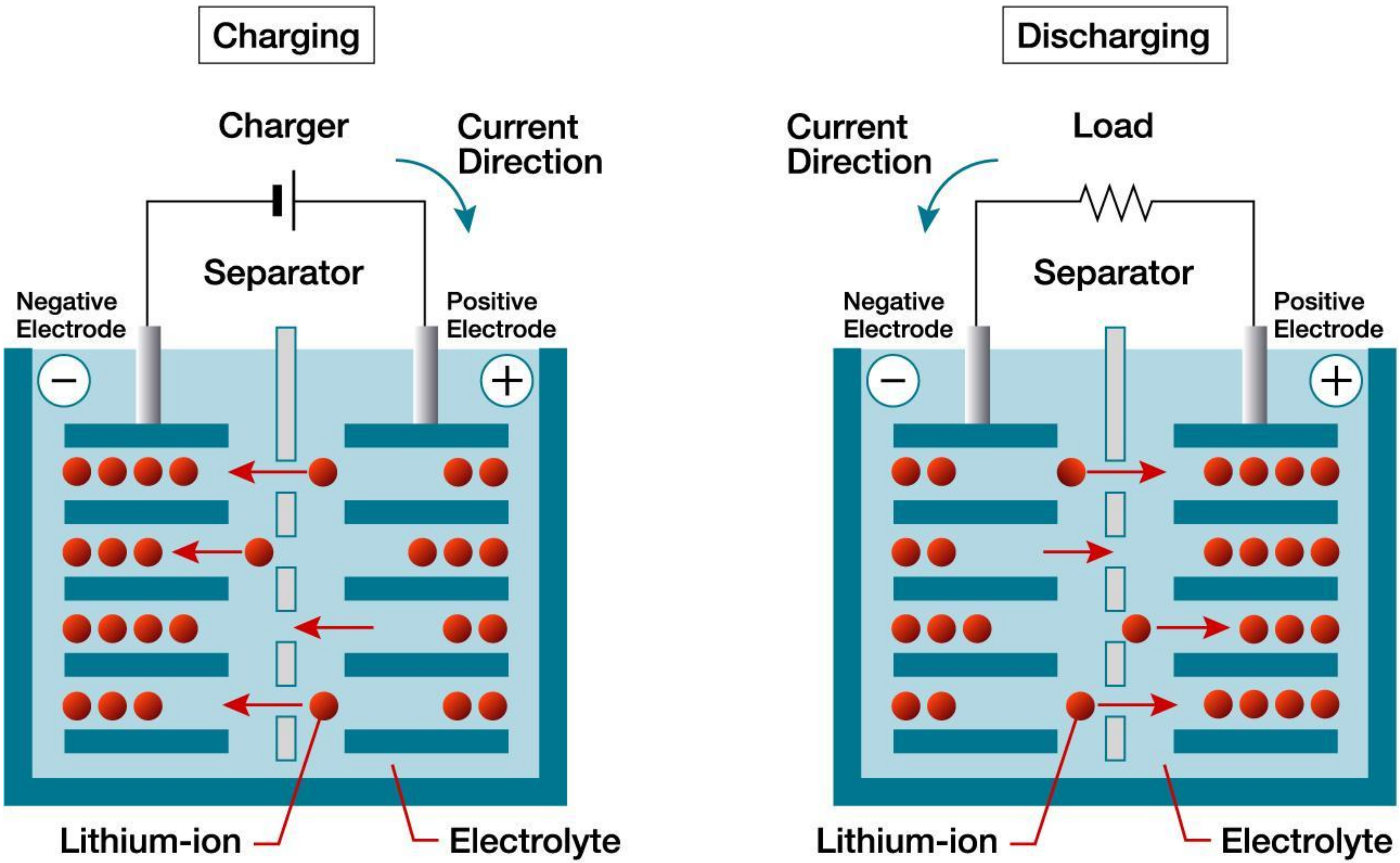
Charging and discharging reactions

Cathode (+): $$LiCoO_2 \underset{discharge}{\overset{charge}{\rightleftharpoons}} Li_{1-x}CoO_2 + xLi^+ + xe^-$$

Anode (-): $$C_n + xLi^+ + xe^- \underset{discharge}{\overset{charge}{\rightleftharpoons}} C_nLi_x$$

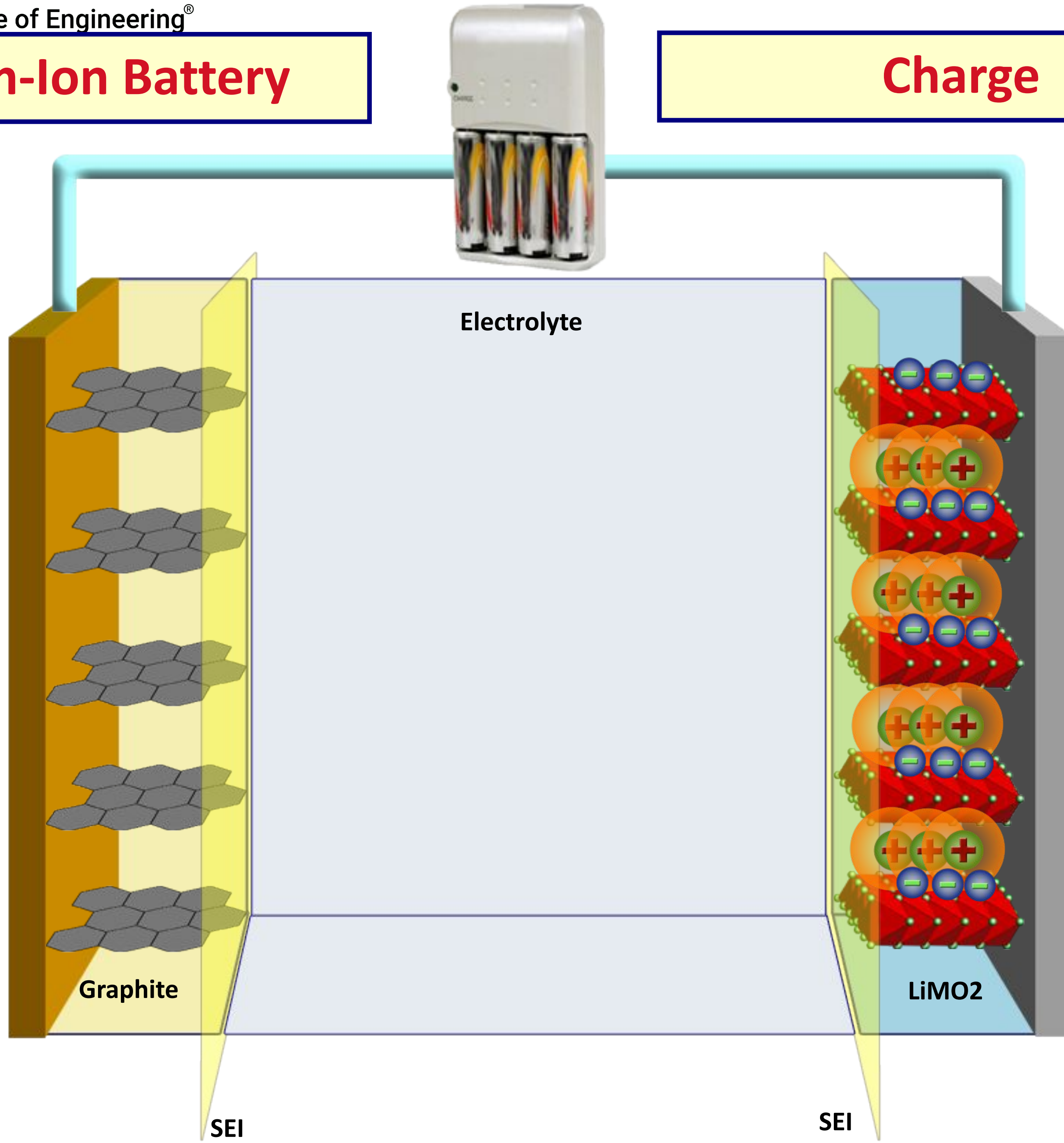
Overall: $$LiCoO_2 + C_n \underset{discharge}{\overset{charge}{\rightleftharpoons}} C_nLi_x + Li_{1-x}CoO_2$$

Charging and discharging reactions



Lithium-Ion Battery

Charge



Charging and discharging reactions

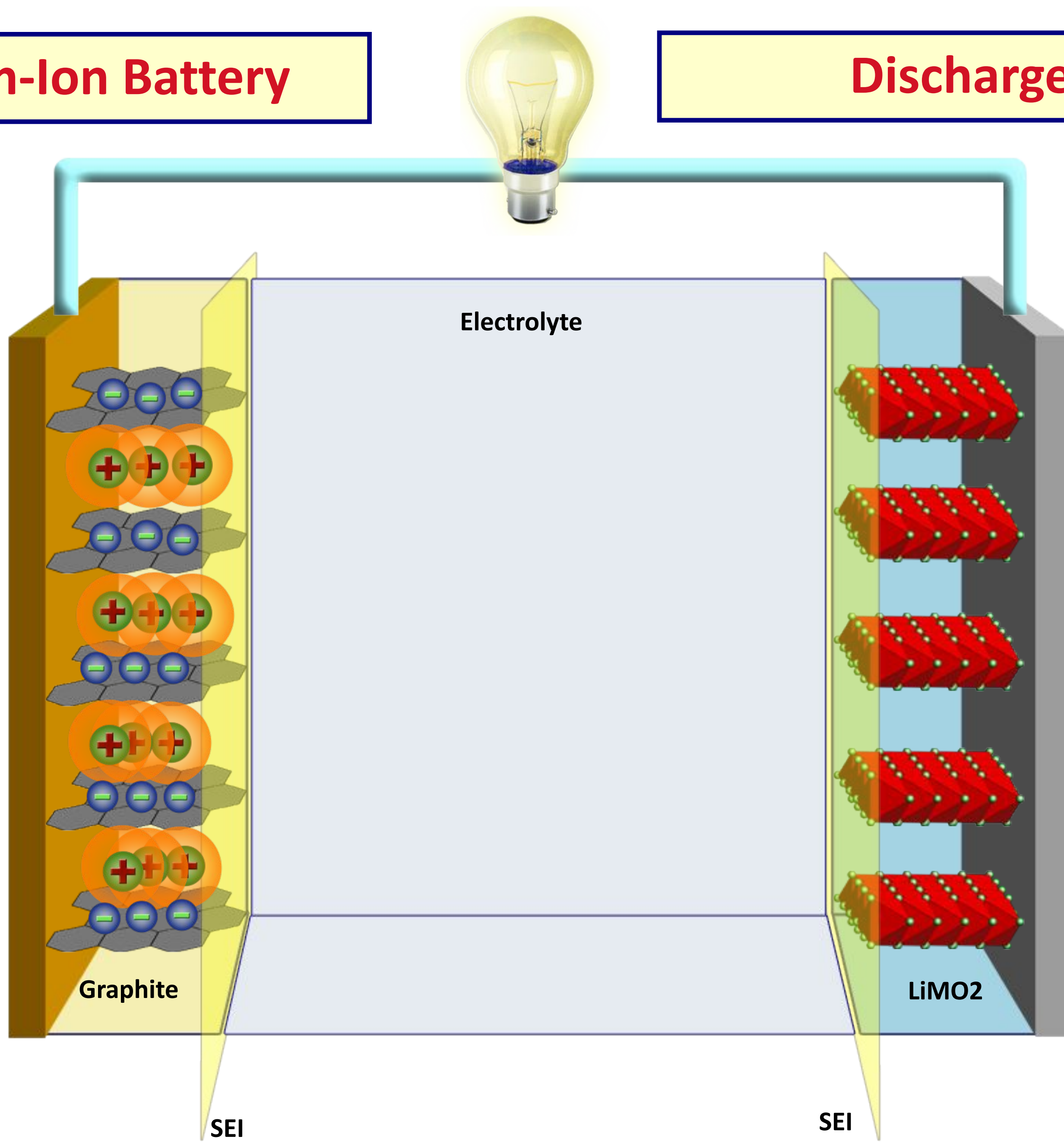
Cathode (+): $LiCoO_2 \underset{discharge}{\overset{charge}{\rightleftarrows}} Li_{1-x}CoO_2 + xLi^+ + xe^-$

Anode (-): $C_n + xLi^+ + xe^- \underset{discharge}{\overset{charge}{\rightleftarrows}} C_nLi_x$

Overall: $LiCoO_2 + C_n \underset{discharge}{\overset{charge}{\rightleftarrows}} C_nLi_x + Li_{1-x}CoO_2$

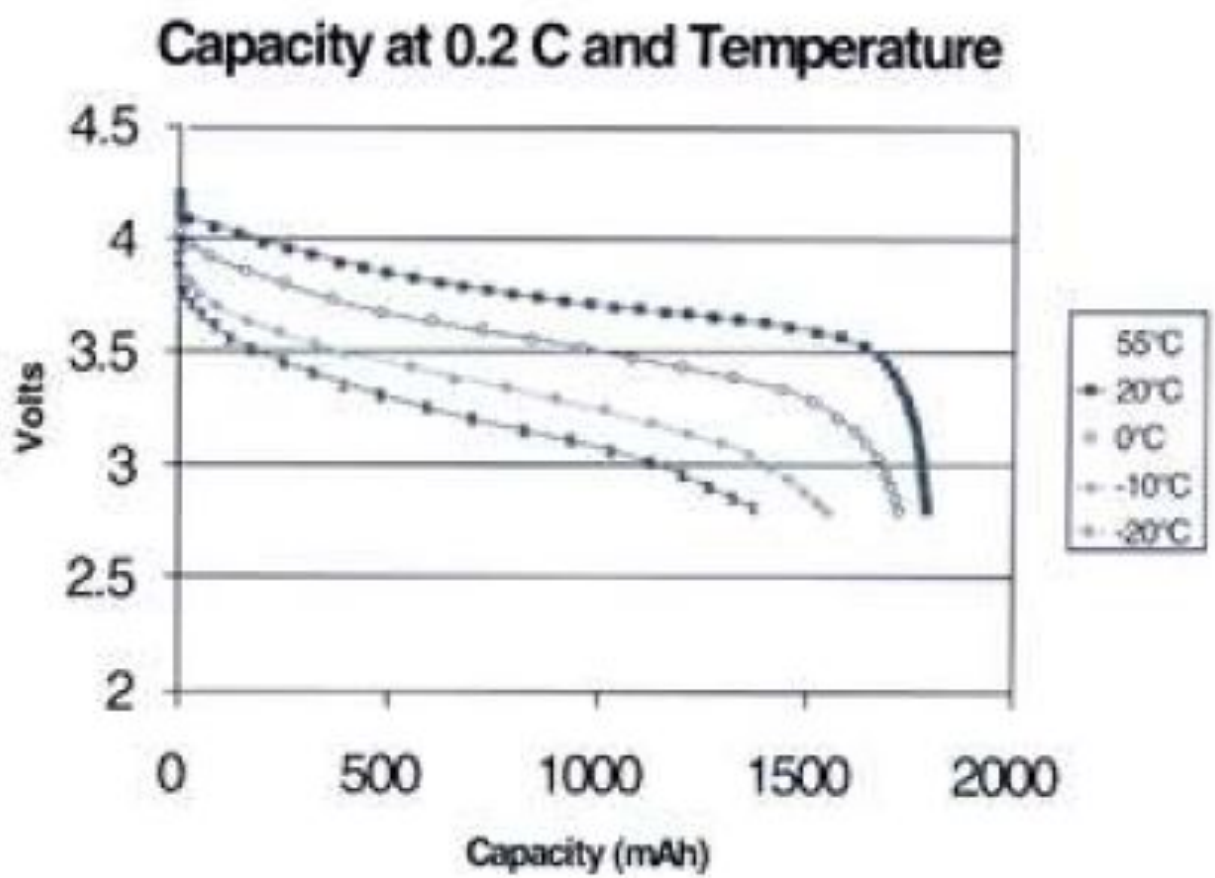
Lithium-Ion Battery

Discharge



Effect of temperature on Li-ion battery performance

- Effects of temperature:

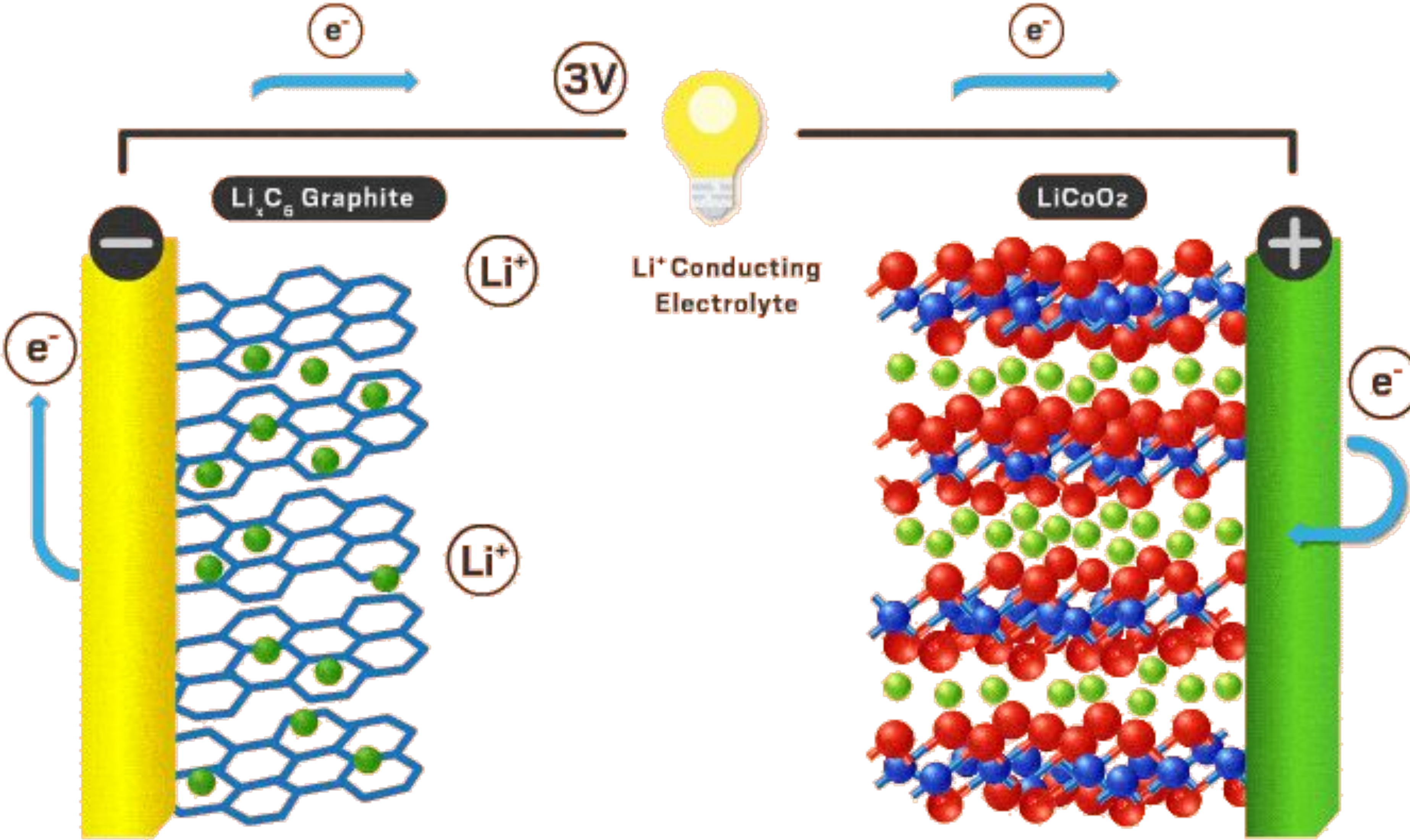


Li-ion batteries

- Advantages with respect to lead-acid batteries:
  - Less sensitive to high temperatures (specially with solid electrolytes)
  - Lighter (compare Li and C with Pb)
  - They do not have deposits every charge/discharge cycle (that's why the efficiency is 99%)
  - Less cells in series are need to achieve some given voltage.
- Disadvantages:
  - Cost

	Lead-Acid	Ni-Cd	Ni-MH	Li-ion
Cell voltage (V)	2	1.2	1.2	3.6
Specific energy (Wh/kg)	1-60	20-55	1-80	3-100
Specific power (W/kg)	< 300	150 – 300	< 200	100 – 1000
Energy density (kWh/m^3)	25-60	25	70-100	80-200
Power density (MW/m^3)	< 0.6	0.125	1.5 – 4	0.4 – 2
Maximum cycles	200-700	500-1000	600-1000	3000
Discharge time range	> 1 min	1 min-8 hr	> 1 min	10 s-1 h
Cost ($/kWh)	125	600	540	600
Cost ($/kW)	200	600	1000	1100
Efficiency (%)	75 - 90	75	81	99

Li-ion batteries (LiCoO2)



Li-ion batteries ($LiCoO_2$)-Pros-cons

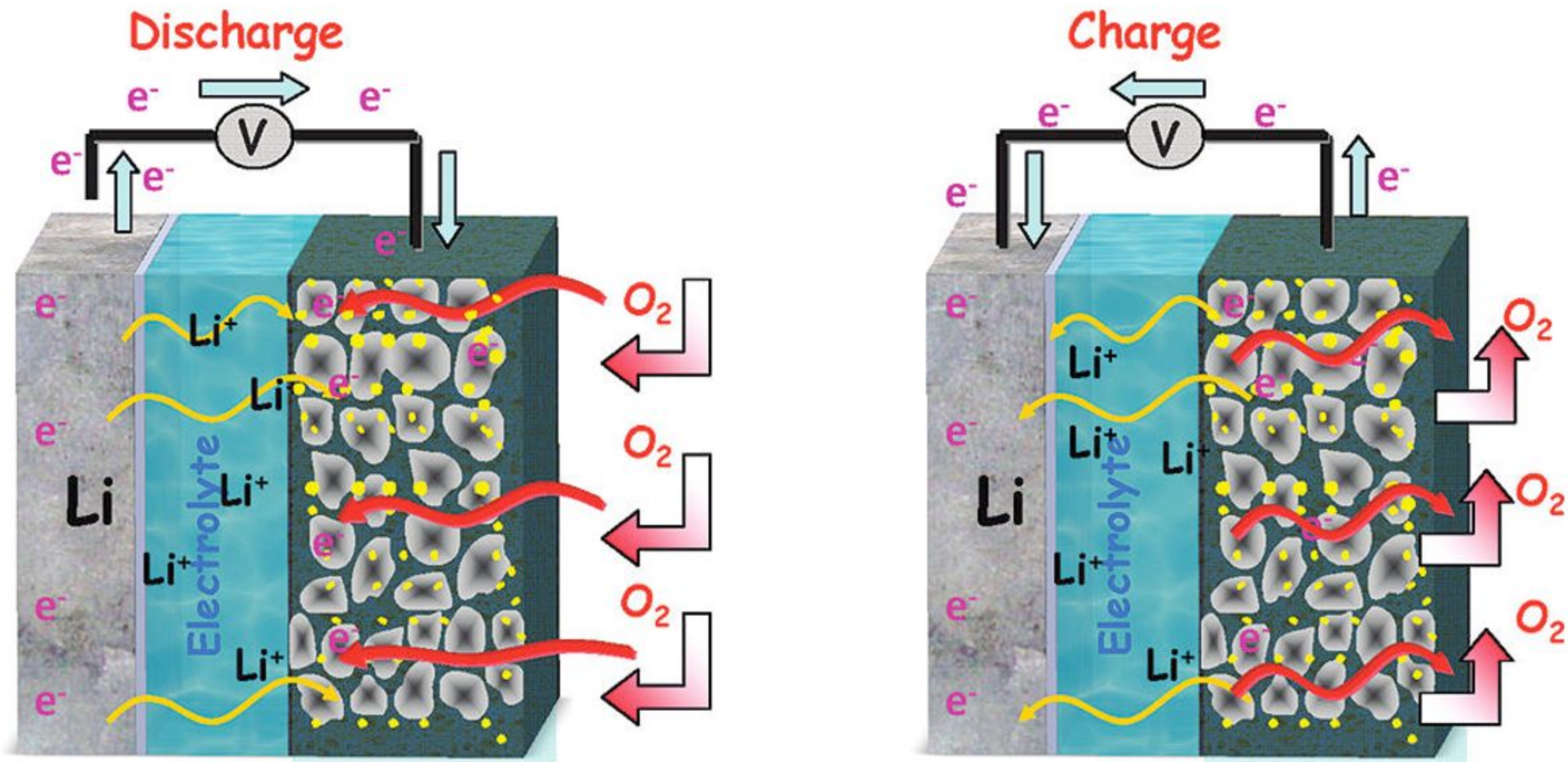
Pros	Cons
Very high energy density	Prone to dendrite formation
Promises rapid energy replenishment	High production costs
Better performance in extremes	Risks of short circuits and safety concerns
Ideal for applications like EVs and drones	Shorter lifespan compared to lithium-ion batteries

Problems on modules and pack

1. In an electric vehicle, battery module developed using lithium ion batteries. Single lithium ion battery has a voltage of 5V with a capacity of 2Ah. Calculate the minimum number of batteries to achieve the characteristic voltage of 40 V and capacity of 100 Ah.

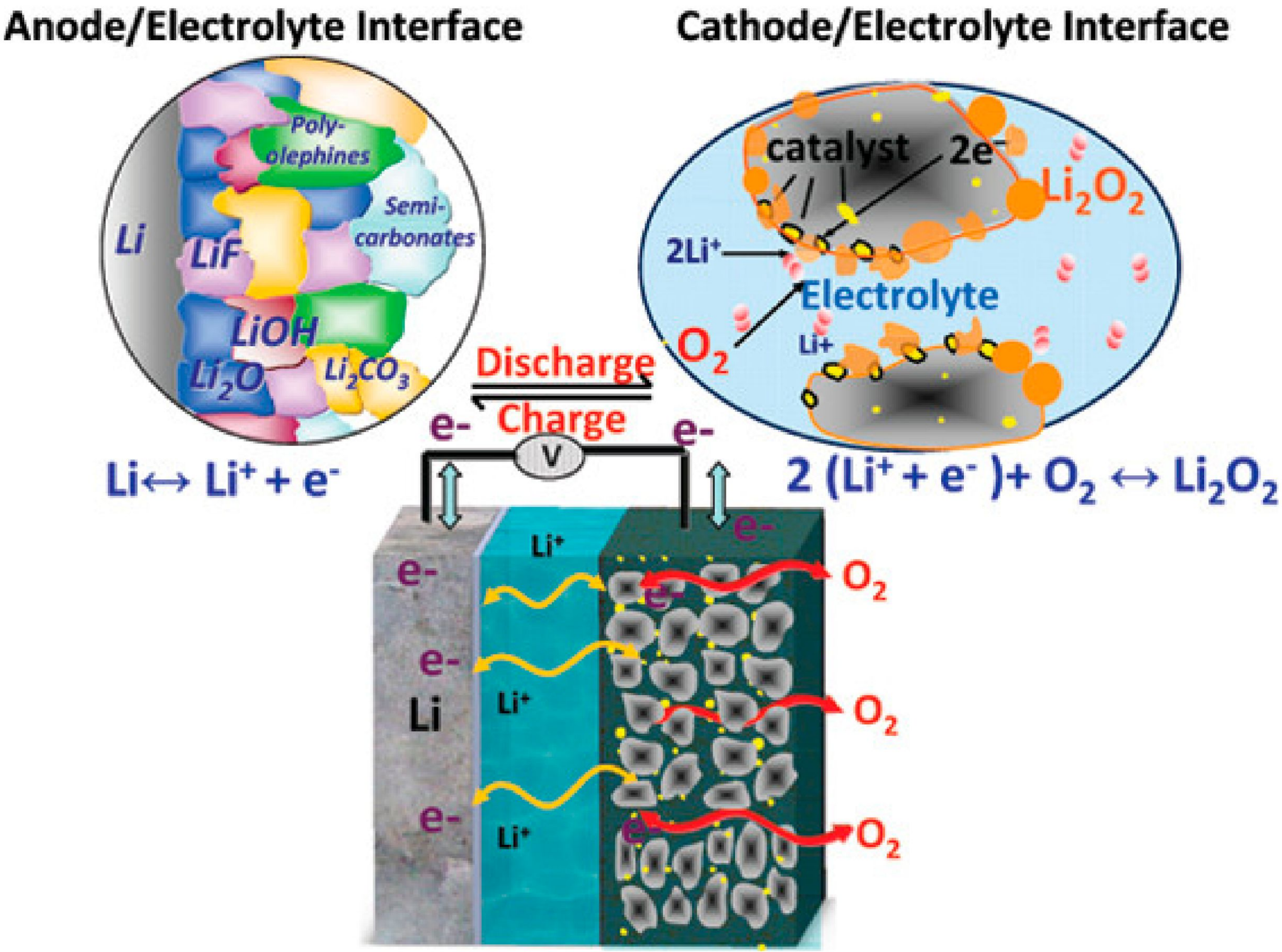
Answer?

Li-air batteries



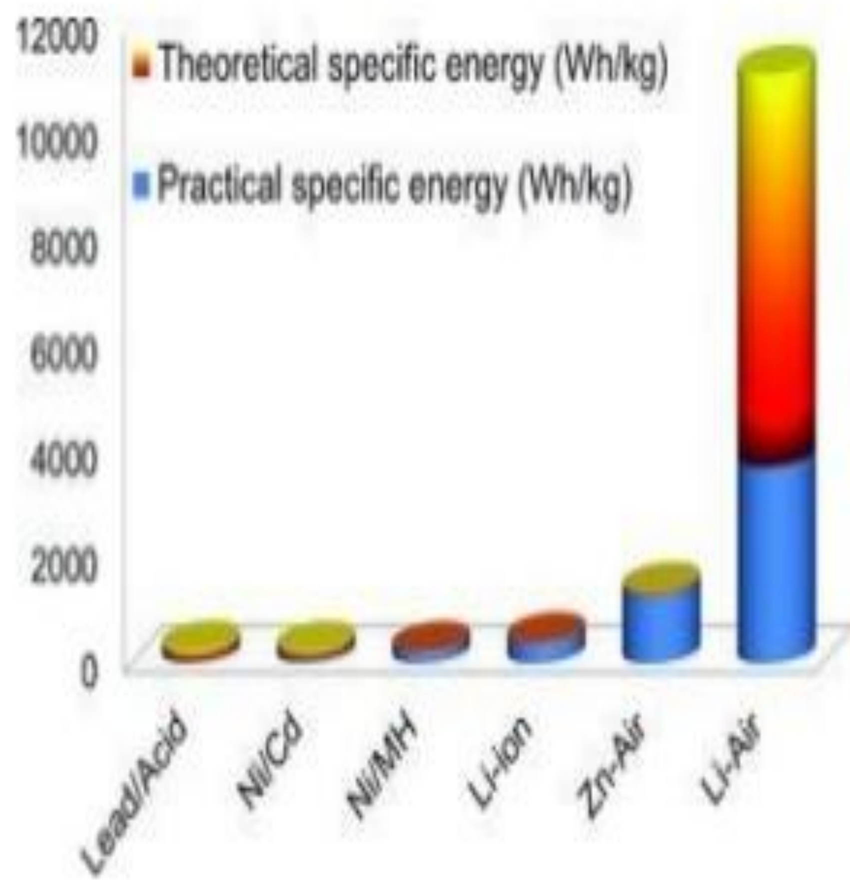
https://www.youtube.com/watch?v=8pMFLpiqPAc

Li-air batteries



Li-air batteries

High specific energy density batteries are attracting growing attention as possible power sources for electric vehicles (EVs).

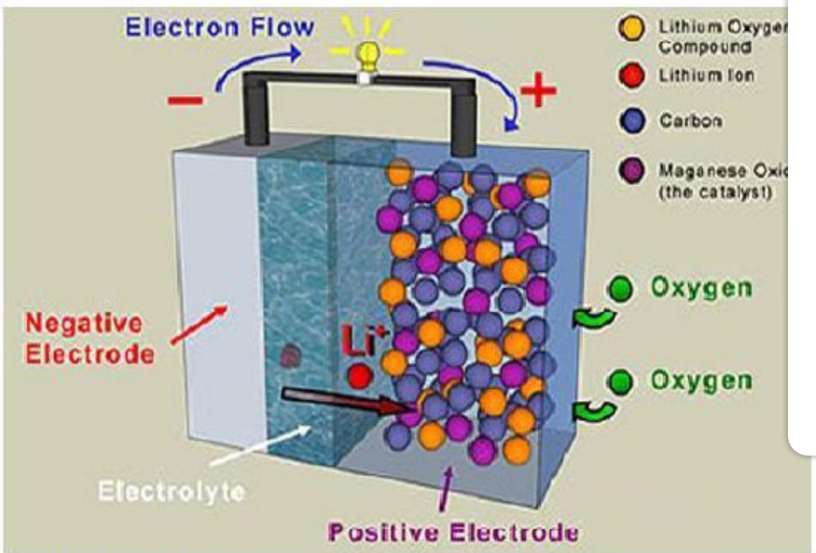


Lithium–air batteries are the most promising system, because of their far higher theoretical specific energy density than conventional batteries.

Li-air batteries

Starting with the knowledge acquired from Li-ion batteries, to obtain batteries with higher energy density, up to 10 fold increase in gravimetric energy density.

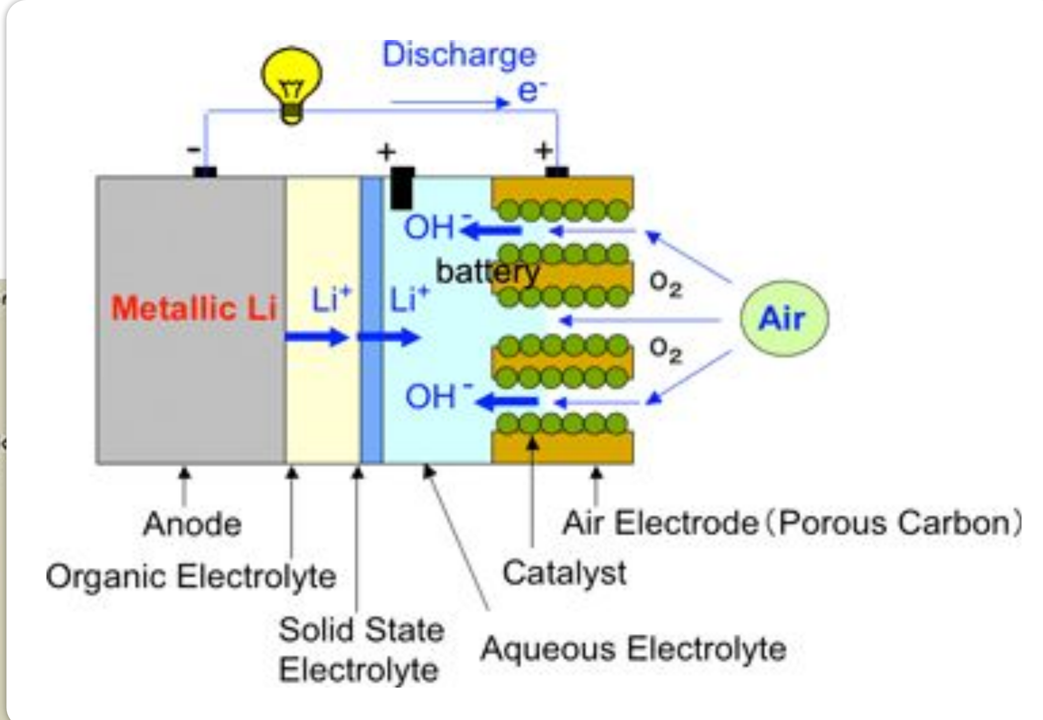
Li-Air BATTERIES



$Li^{(s)} \leftrightarrow Li^+ + e^-$ (anode)

$Li^+ + 1/2O_2 + e^- \leftrightarrow 1/2Li_2O_2$ (cathode)

$Li^+ + 1/4O_2 + e^- \leftrightarrow 1/2Li_2O$ (cathode)



- Discharge voltage of 2.7 V
- Theoretically energy density: >11500 Wh/kg based on Li only

Question: practically, can it really compete with Li-ion and what are main issues?

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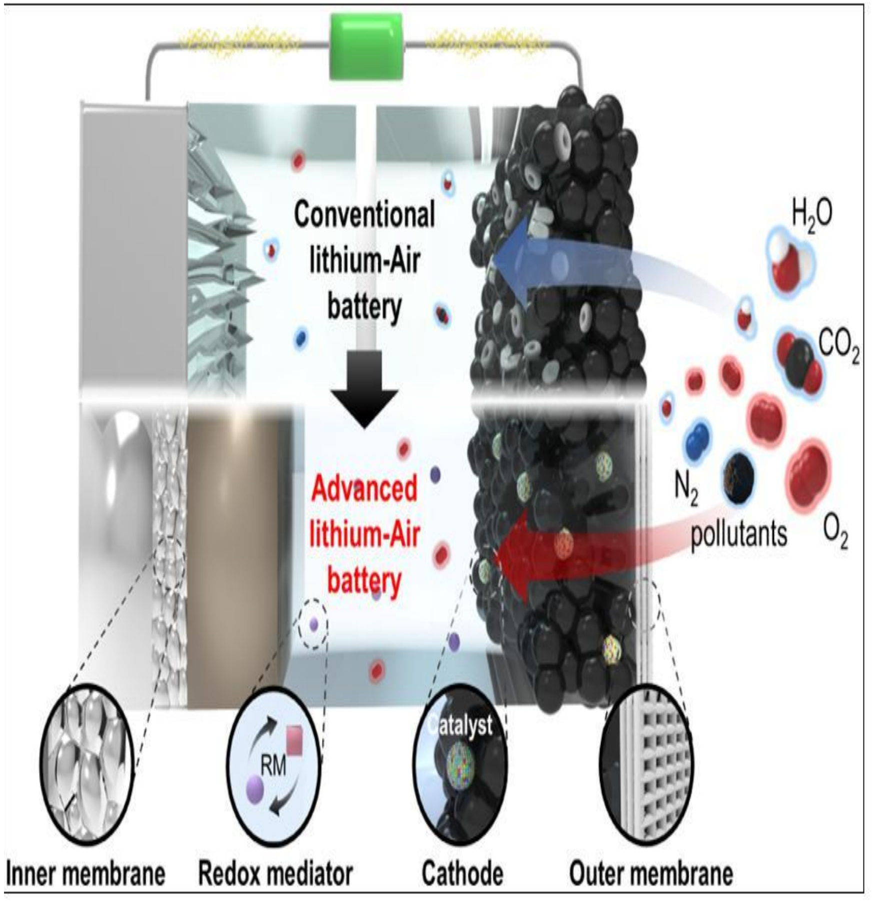
Li-air batteries: introduction

- The lithium-air battery (Li-air) is a metal–air electrochemical cell.

- It works by oxidation of lithium at the anode and reduction of oxygen at the cathode to induce a current flow.

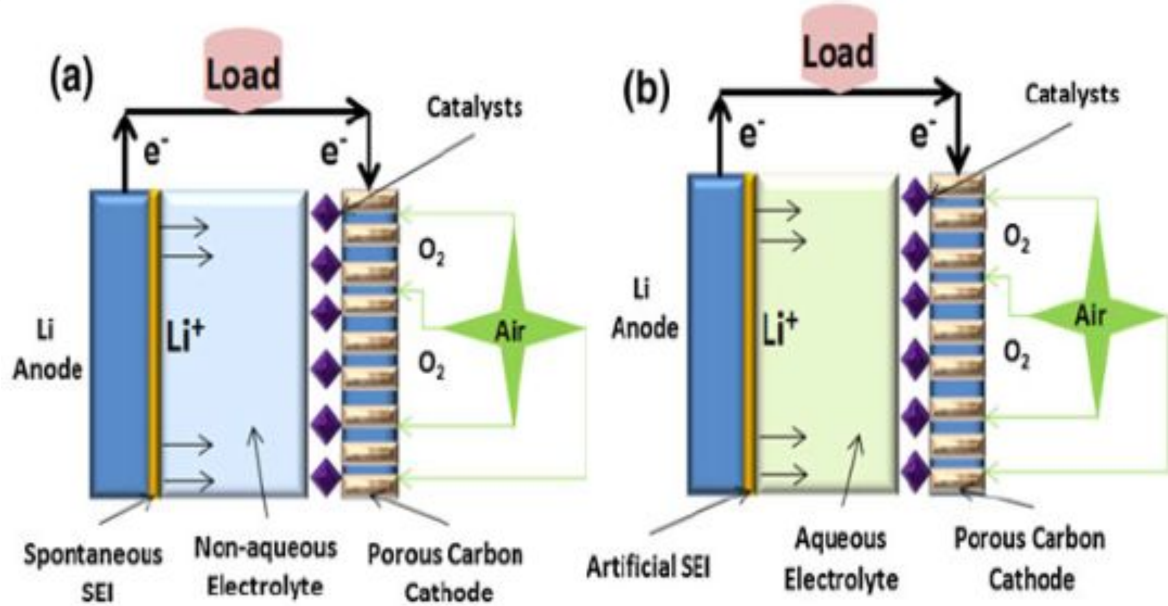
- A metal–air electrochemical cell is an electrochemical cell that uses an anode made from pure metal and an external cathode of ambient air, typically with an aqueous or aprotic electrolyte.

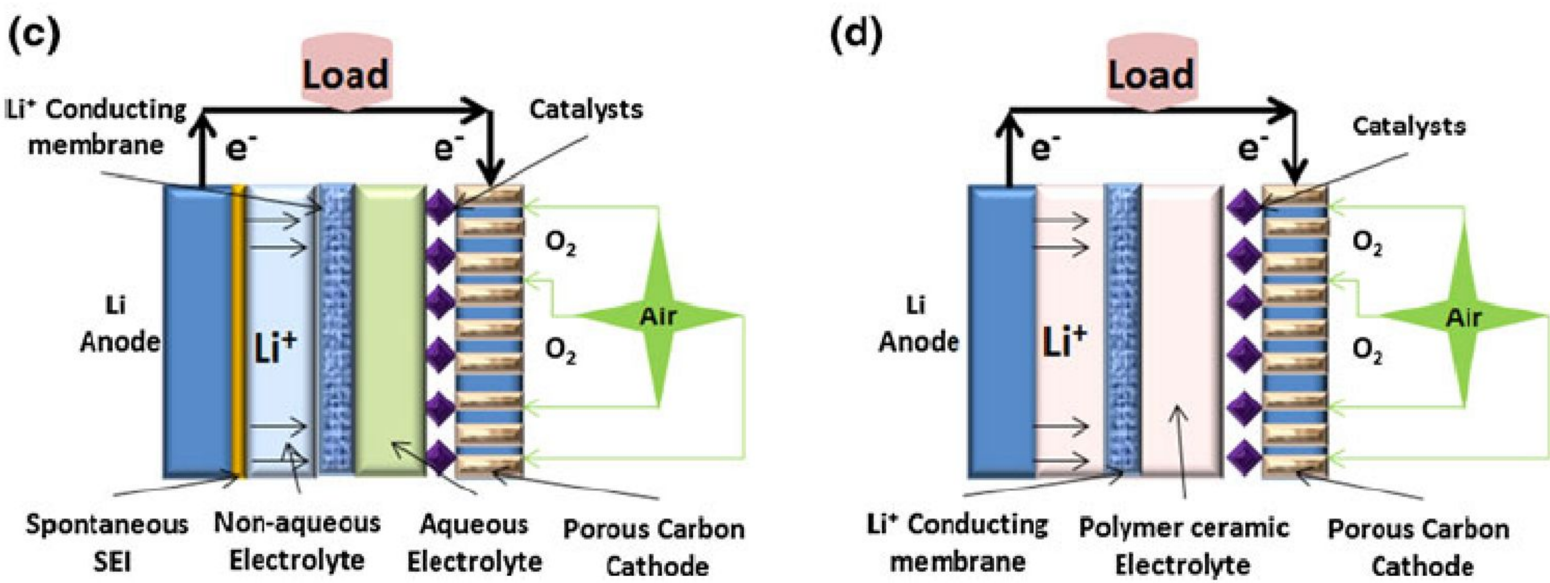


Types of Li-air batteries

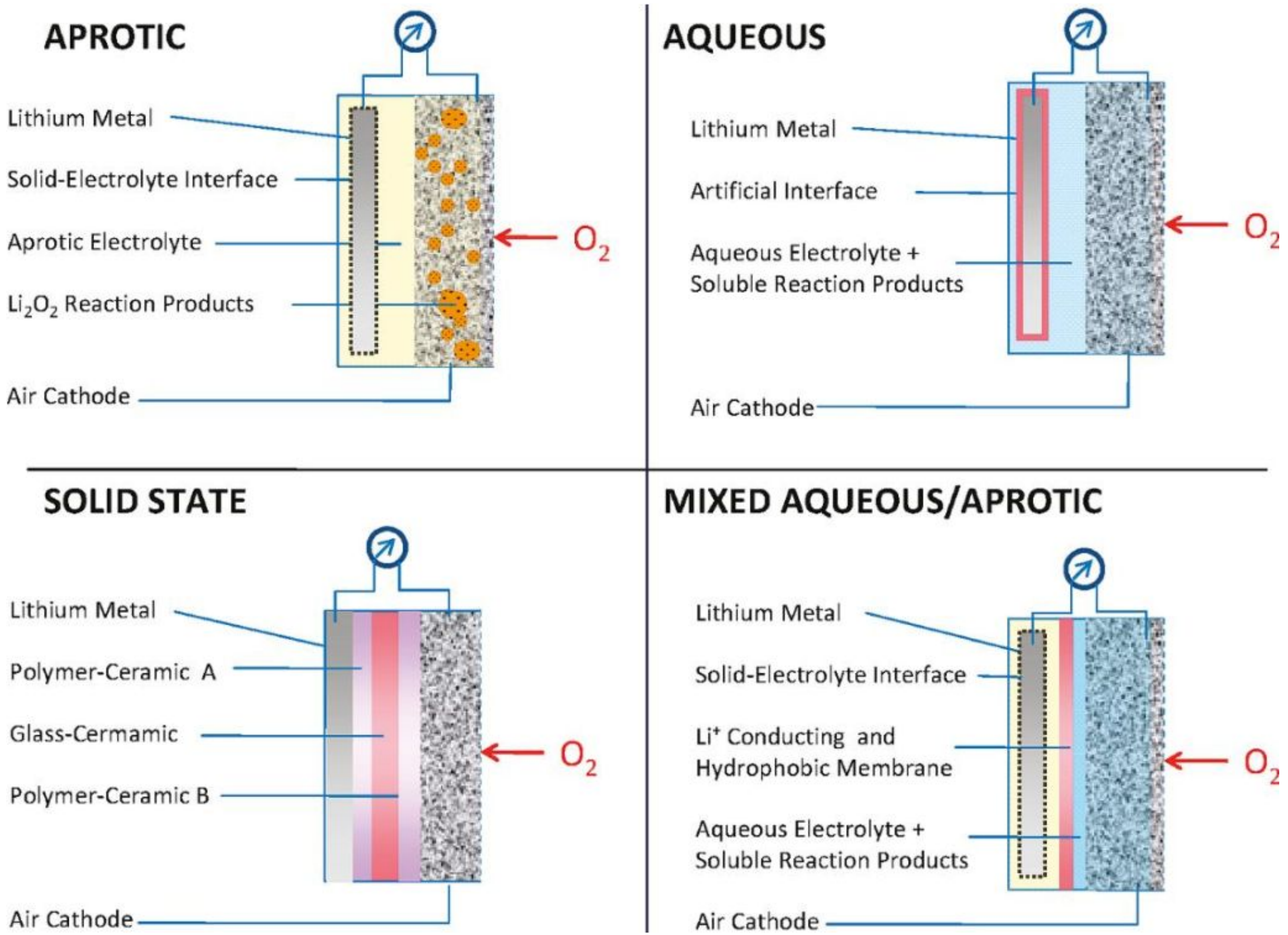
Lithium air batteries based on the electrolyte used can be classified into four types:

a) Fully Aprotic (Non-aqueous electrolytes)
b) Aqueous electrolytes
c) Hybrid electrolytes [Combination of (a) & (b or (d)]
d) Solid state electrolytes



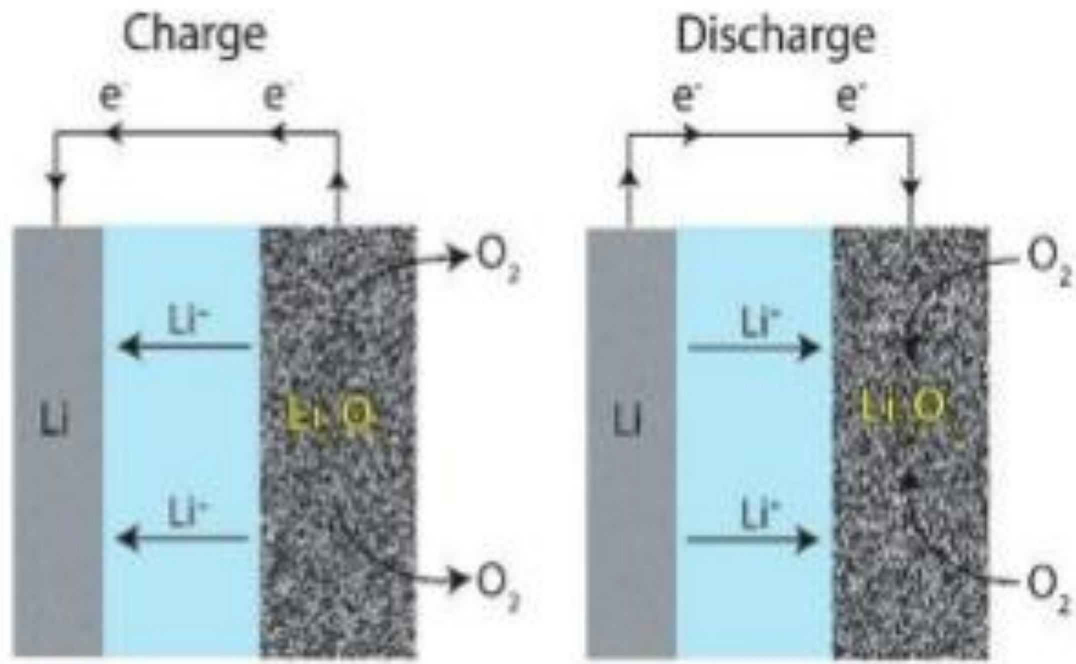


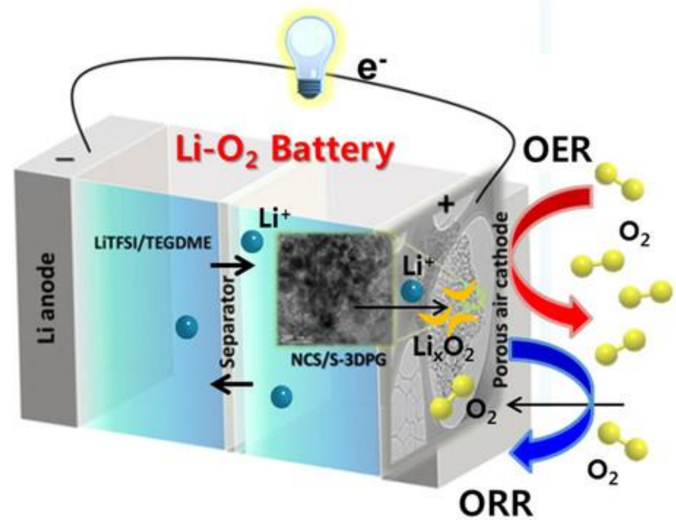
Types of Li-air batteries



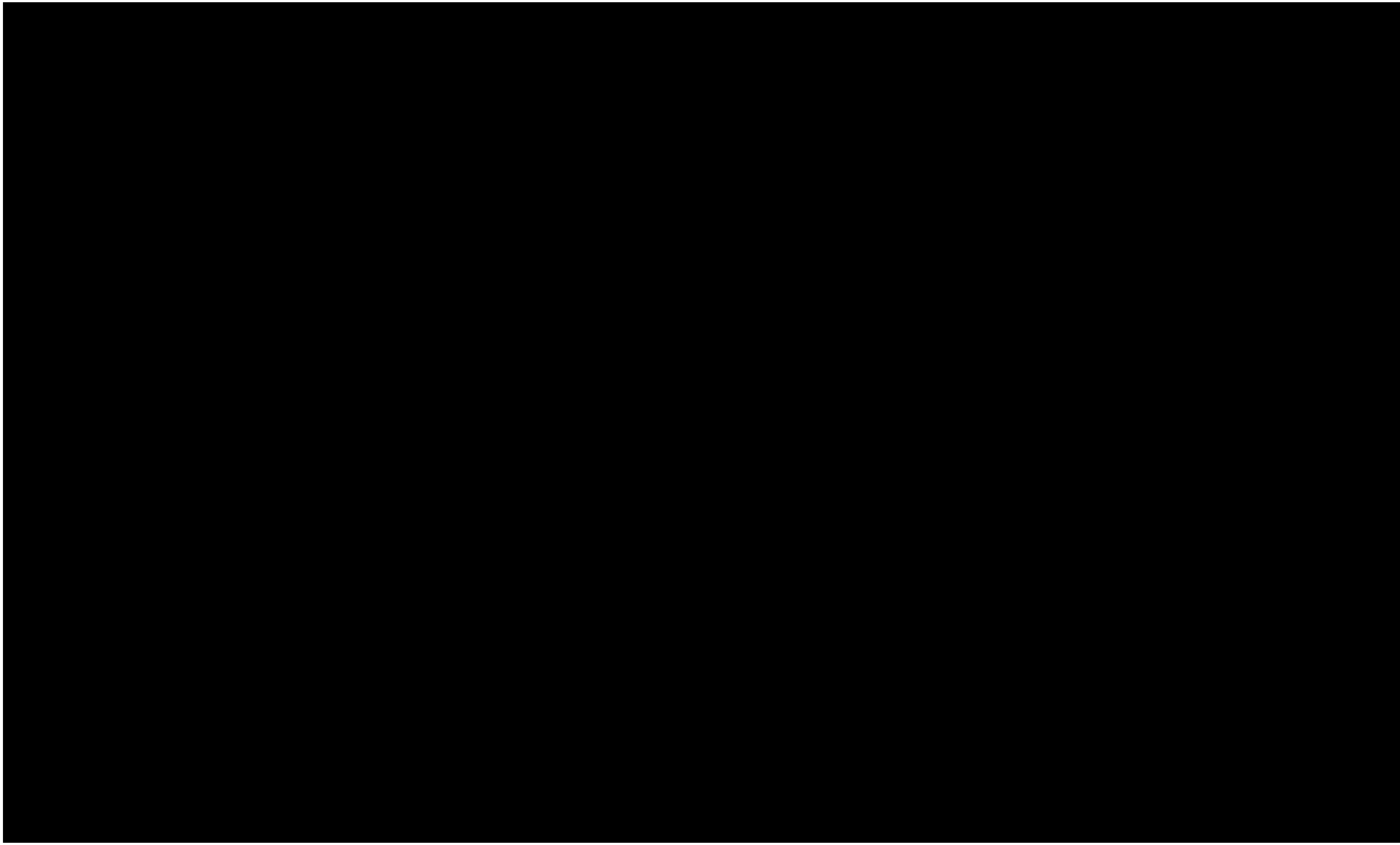
Working

- In general lithium ions move between the anode and the cathode across the electrolyte.
- Under discharge, electrons follow the external circuit to do electric work and the lithium ions migrate to the cathode.
- During charge the lithium metal plates onto the anode, freeing O_2 at the cathode.
- Both non-aqueous (with Li_2O_2 or LiO_2 as the discharge products) and aqueous (LiOH as the discharge product) Li-O_2 batteries have been considered.
- The aqueous battery requires a protective layer on the negative electrode to keep the Li metal from reacting with water.





Working



Anode material

- Lithium metal is the typical anode choice.
- At the anode, electrochemical potential forces the lithium metal to release electrons via oxidation (without involving the cathodic oxygen).
- The half-reaction is

$$Li \leftrightarrow Li^{+} + e^{-}$$

- Upon charging/discharging in aprotic cells, layers of lithium salts precipitate onto the anode, eventually covering it and creating a barrier between the lithium and electrolyte.
- This barrier initially prevents corrosion, but eventually inhibits the reaction kinetics between the anode and the electrolyte.

Cathode material

- At the cathode during charge, oxygen donates electrons to the lithium via reduction.
- Mesoporous carbon has been used as a cathode substrate with metal catalysts that enhance reduction kinetics and increase the cathode's specific capacity.
- Manganese, cobalt, ruthenium, platinum, silver, or a mixture of cobalt and manganese are potential metal catalysts.
- In a cell with an aprotic electrolyte lithium oxides are produced through reduction at the cathode:

$$Li^+ + e^- + O_2 + * \rightarrow LiO_2^*$$

$$Li^+ + e^- + LiO_2^* \rightarrow Li_2O_2^*$$

- where "*" denotes a surface site on Li_2O_2 where growth proceeds, which is essentially a neutral Li vacancy in the Li_2O_2 surface.

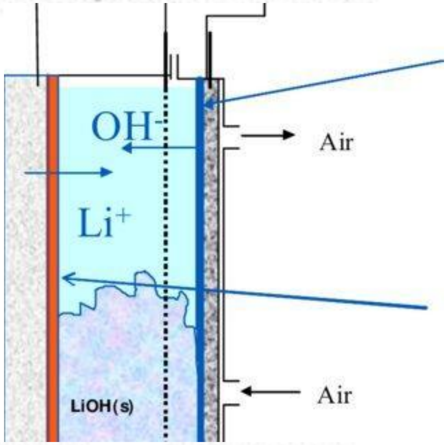
Electrolyte

In a cell with an aqueous electrolyte the reduction at the cathode can also produce lithium hydroxide:

- **Acidic electrolyte**

$$2Li + \frac{1}{2}O_2 + 2H^+ \rightarrow 2Li^+ + H_2O$$

  - A conjugate base is involved in the reaction.
  - The theoretical maximal Li-air cell specific energy and energy density are 1400 W·h/kg and 1680 W·h/l, respectively.



- **Alkaline aqueous electrolyte**

$$2Li + \frac{1}{2}O_2 + H_2O \rightarrow 2LiOH$$

  - Water molecules are involved in the redox reactions at the air cathode.
  - The theoretical maximal Li-air cell specific energy and energy density are 1300 W·h/kg and 1520 W·h/l, respectively

Aprotic(non aqueous)Li-air batteries

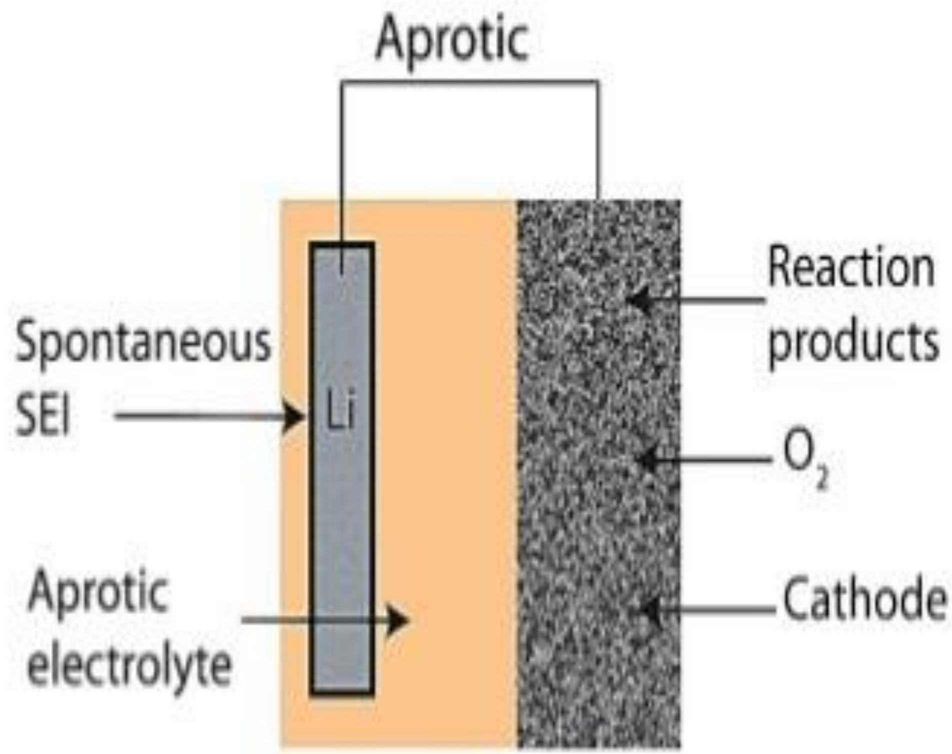
- Several chemical products may result from the reaction of Li with O_2, depending on the chemical environment and mode of operation.
- Most effort involved aprotic materials, which consist of a lithium metal anode, a liquid organic electrolyte and a porous carbon cathode.
- The electrolyte can be made of any organic liquid able to solvate lithium salts such as $LiPF_6$, $LiAsF_6$, $LiN(SO_2CF_3)_2$, and $LiSO_3CF_3$), but typically consisted of carbonates, ethers and esters.
- Most studies agree that Li_2O_2 is the final discharge product of non-aqueous Li-O_2 batteries.
- In nonaqueous Li/air batteries there are two principal electrode reactions of interest:

$$2Li + \frac{1}{2}O_2 \leftrightarrow Li_2O \quad \& \quad 2Li + O_2 \leftrightarrow Li_2O_2$$

- In the absence of practical considerations the full reduction of O_2 to Li_2O is desired because of its higher specific energy and energy density, but it appears that Li_2O_2 is a product that forms more readily than Li_2O.
- In addition, when Li_2O_2 is formed full cleavage of the O-O bond may not be necessary, which is important from a kinetic point of view.

Aprotic(non aqueous)Li-air batteries

- The open circuit voltage (OCV) of 2.96 V, are 3460 Wh kg1 and 6940 Wh L1 for the discharge state.
- For the charged state, oxygen is excluded and the specific energy density is 11,680 Wh kg1 which is close to the energy density of gasoline (ca. 13,000 Wh kg1).
- The energy density per unit mass and per unit volume of non-aqueous lithium–air batteries is about 10 times and 6 times higher than those of lithium-ion batteries, respectively, with a carbon anode and $LiCoO_2$ cathode.

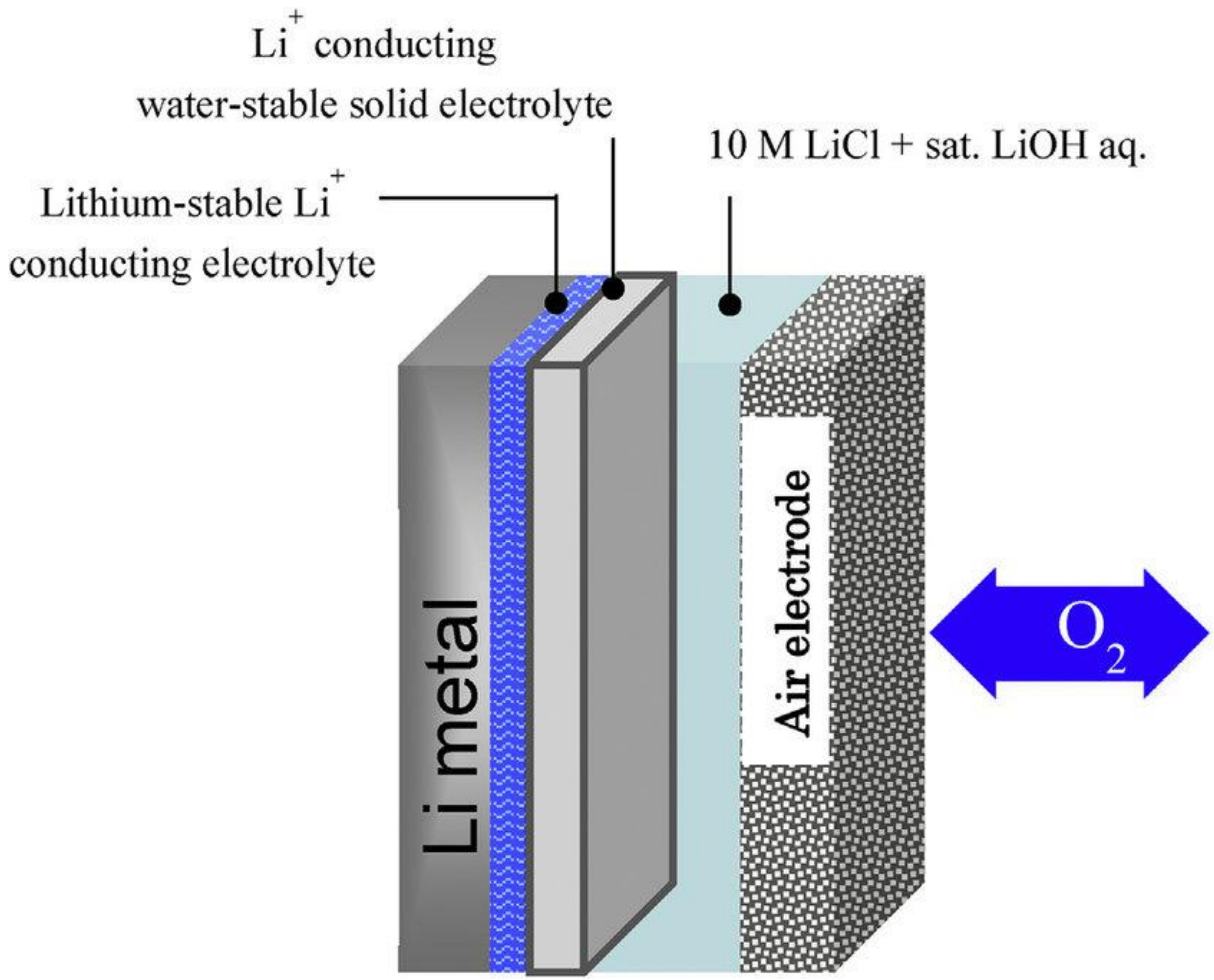


Schematic of aprotic type Li-Air battery design

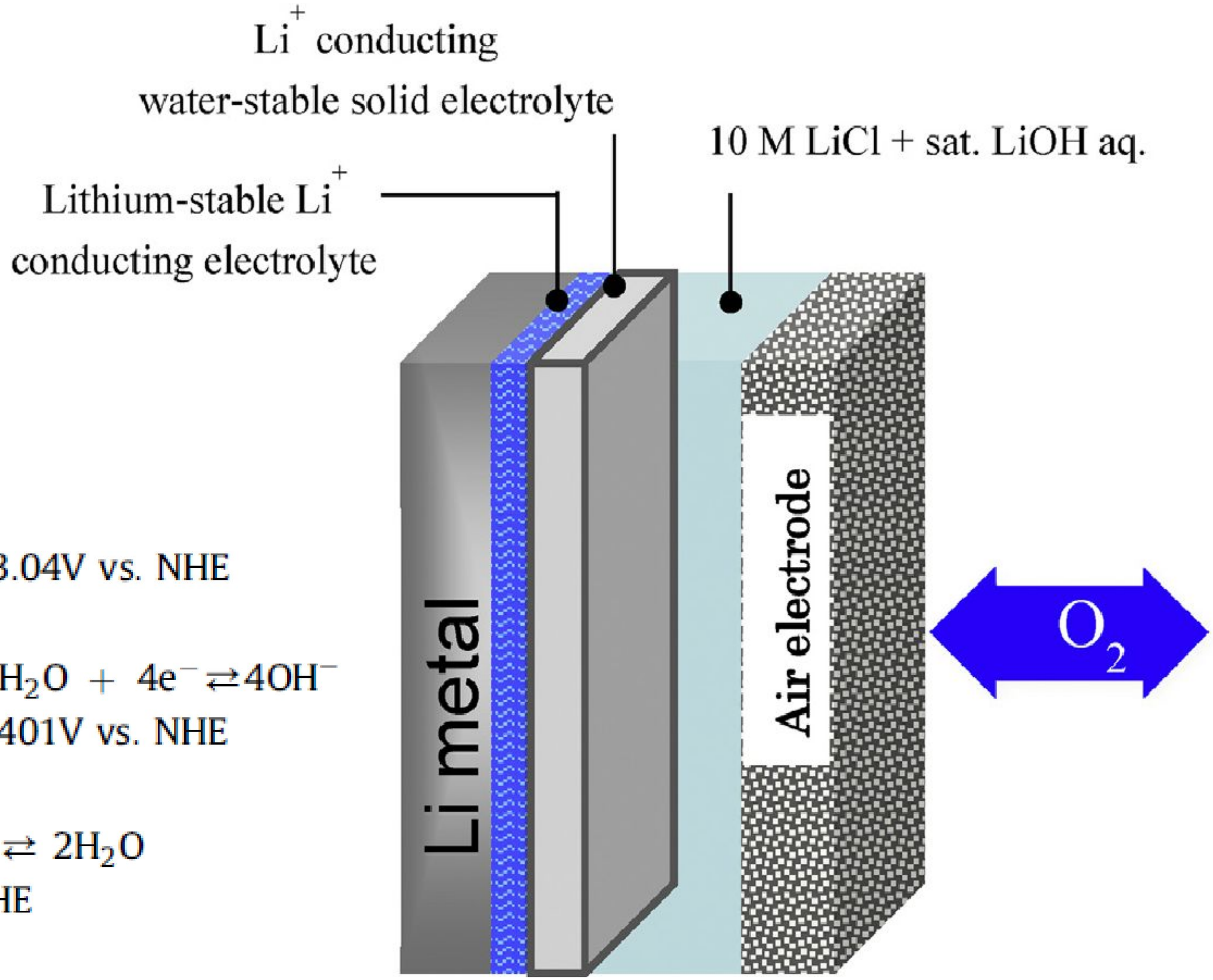
Aqueous Li-air batteries

- An aqueous Li-air battery consists of a lithium metal anode, an aqueous electrolyte and a porous carbon cathode.
- The aqueous electrolyte combines lithium salts dissolved in water.
- It avoids the issue of cathode clogging because the reaction products are water-soluble.
- The aqueous design has a higher practical discharge potential than its aprotic counterpart.
- However, lithium metal reacts violently with water and thus the aqueous design requires a solid electrolyte interface between the lithium and electrolyte.
- Commonly, a lithium-conducting ceramic or glass is used, but conductivity are generally low (on the order of 10^{-3} S/cm at ambient temperatures).

Aqueous Li-air batteries



Aqueous Li-air batteries



anode: $Li \rightleftarrows Li^+ + e^-$ $E^\circ = -3.04V$ vs. NHE

cathode: (alkaline) $O_2 + 2H_2O + 4e^- \rightleftarrows 4OH^-$
$E^\circ = 0.401V$ vs. NHE

(acid) $O_2 + 4H^+ + 4e^- \rightleftarrows 2H_2O$
$E^\circ = 1.229V$ vs. NHE

Aqueous Li-air batteries mechanism

- Reactions involving Li and O_2 in an aqueous medium depend on the pH.

- In a basic aqueous environment O_2 reduction includes H_2O as a reactant and results in the formation of LiOH

$$2Li + \frac{1}{2}O_2 + H_2O \leftrightarrow 2LiOH \qquad (1)$$

- The product of this reaction is aqueous LiOH, which has a solubility limit of about 5.25 M at standard temperature and pressure.

- If LiOH exceeds its solubility limit it will precipitate out of the solution as a monohydrate, $LiOH.H_2O$, rather than LiOH.

$$4Li+6H_2O+O_2 = 4(LiOH.H_2O) \qquad (2)$$

- This is a critical point for calculating the specific energy and energy density of aqueous Li/air batteries.

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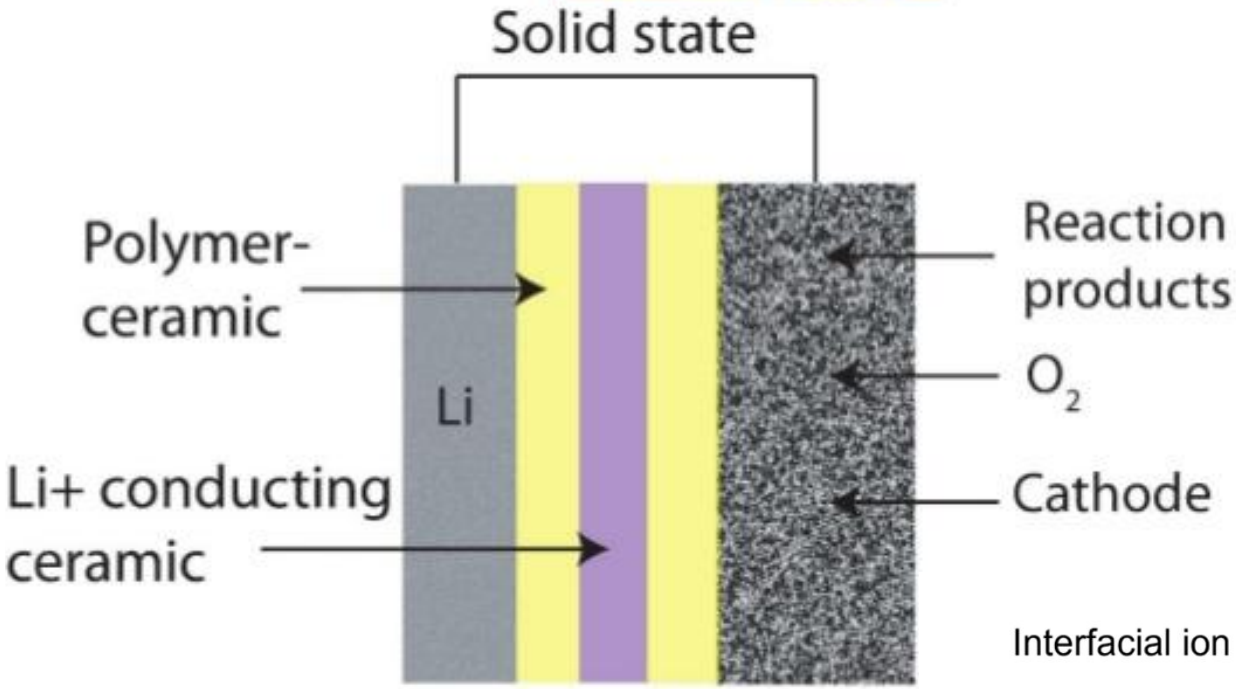
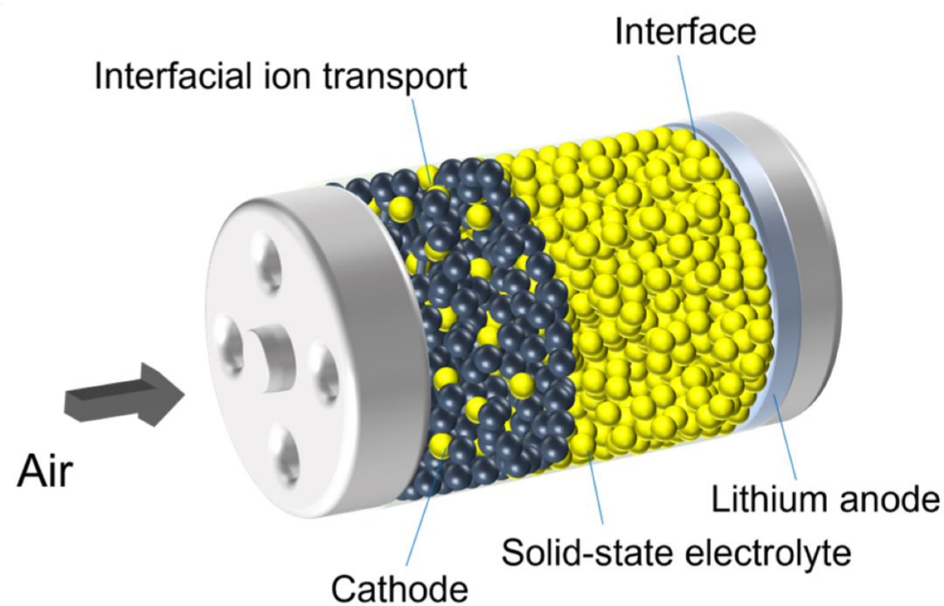
Solid state Li-air batteries

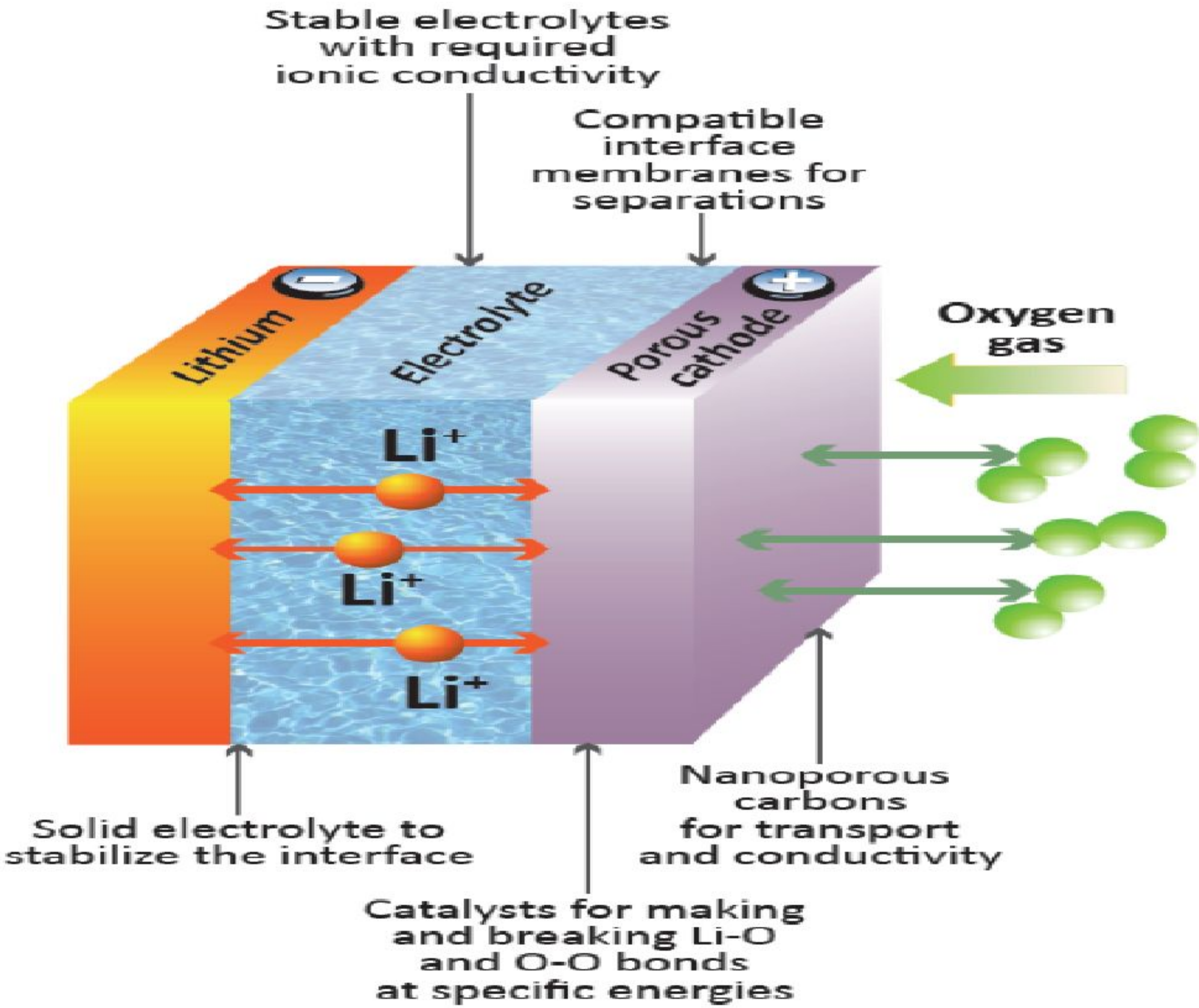
- A solid-state battery design is attractive for its safety, eliminating the chance of ignition from rupture.
- Current solid-state Li-air batteries use a lithium anode, a ceramic, glass, or glass-ceramic electrolyte, and a porous carbon cathode.
- The anode and cathode are typically separated from the electrolyte by polymer-ceramic composites that enhance charge transfer at the anode and electrochemically couple the cathode to the electrolyte.
- The polymer-ceramic composites reduce overall impedance.
- The main drawback of the solid-state battery design is the low conductivity of most glass-ceramic electrolytes.
- The ionic conductivity of current lithium fast ion conductors is lower than liquid electrolyte alternatives.

Solid state Li-air batteries

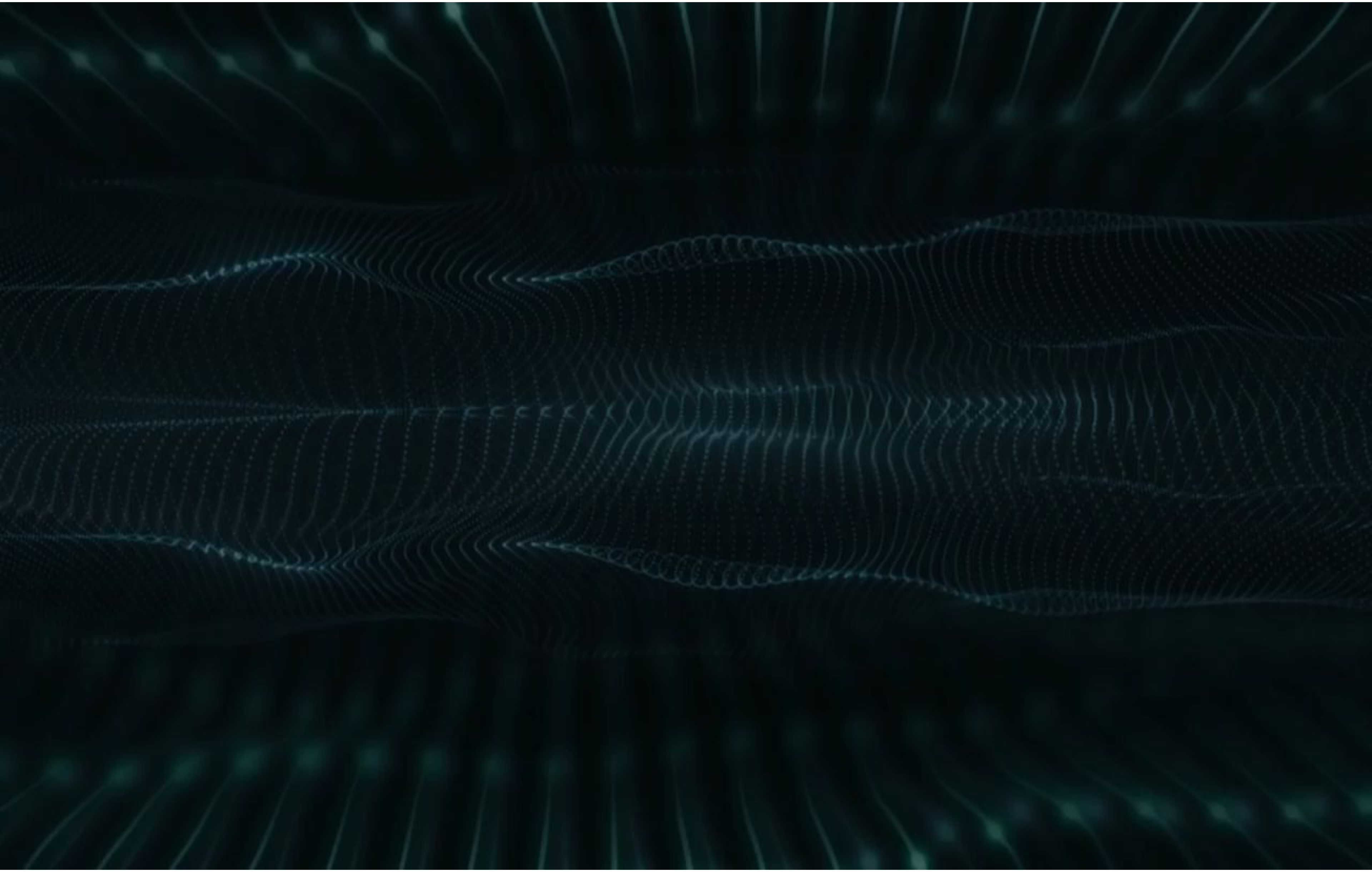




Solid state Li-air batteries



Solid state Li-air batteries



Advantages of Solid state Li-air batteries

Simple structure → Higher energy density; No cost for separator

- Solid electrolytes light play a role as a separator preventing contact of the anode and cathode

High voltage → Higher energy density

- The decomposition voltage of solid electrolytes is high

Non-flammable solid electrolyte → Safety; Larger operating temperature range; Higher energy density

- Can be used at higher operating temperature
- Electrolyte is flame-retardant
- No liquid electrolyte leak risk
- Possibility of cell stacking in one package
- Less or simpler safety measures needed on the pack level

Simpler cell structure and simpler manufacturing process* → Lower costs

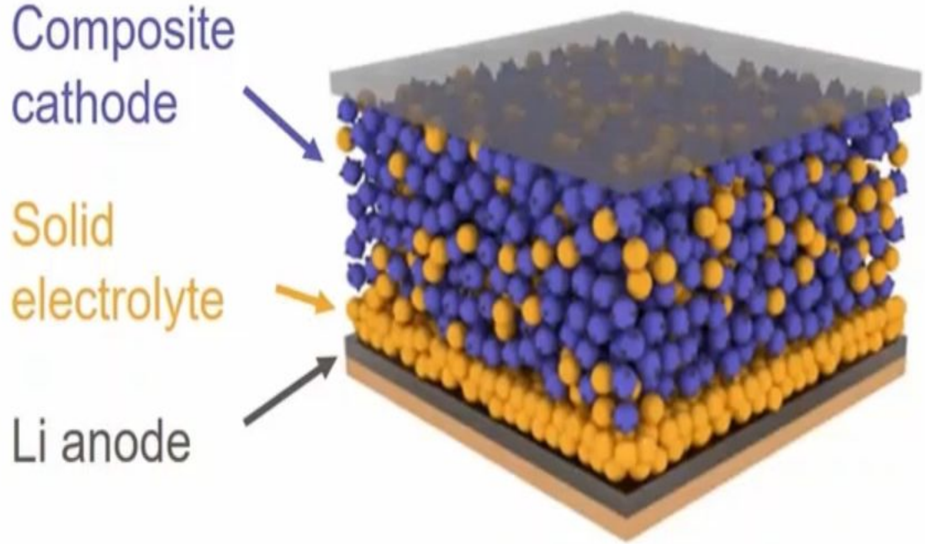
Solid state Li-air batteries

Advantages:

- Safety
- Power density ($t_+ = 1$)
- Enable Li anode
- Longer life

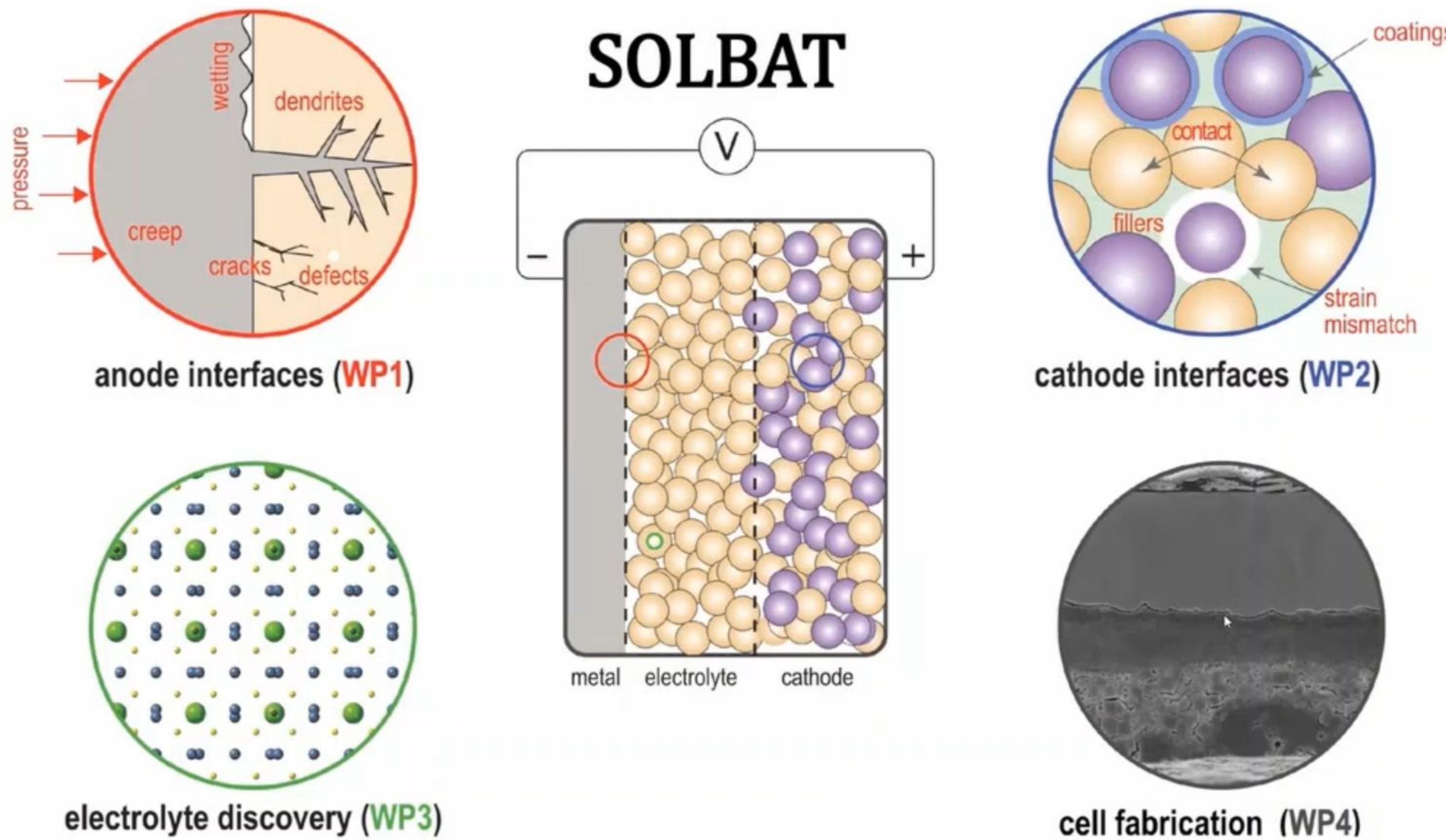
Disadvantages:

- Unproven
- Cost
- Manufacturability
- Electrochemo-mechanics



Energy Density	Current Li-ion	Projected SSBs
Gravimetric (Wh/kg)	250	400
Volumetric (Wh/L)	700	1200

Solid state Li-air batteries



Solid state Li-air batteries: Challenges

Cathode

- Incomplete discharge due to blockage of the porous carbon cathode with discharge products such as lithium peroxide (in aprotic designs) is the most serious.

- The effects of pore size and pore size distribution remain poorly understood.

- Catalysts have shown promise in creating preferential nucleation of Li2O2 over Li2O, which is irreversible with respect to lithium.

- Atmospheric oxygen must be present at the cathode, but contaminants such as water vapor can damage it.

Solid state Li-air batteries: Challenges

Stability

- Long-term battery operation requires chemical stability of all cell components.

- Current cell designs show poor resistance to oxidation by reaction products and intermediates.

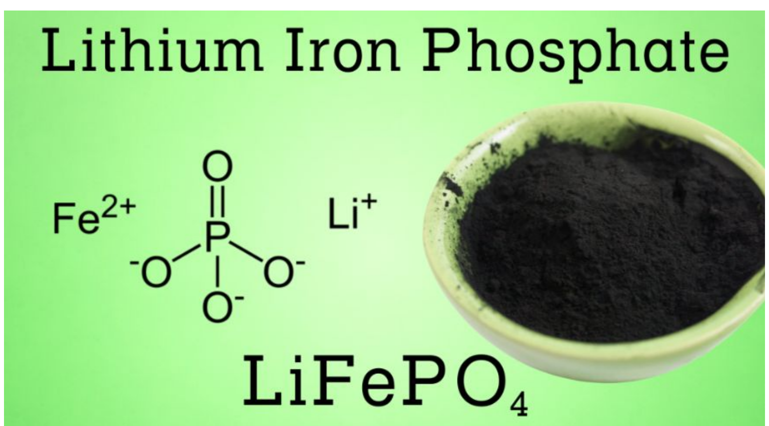
- Stability is hampered in general by parasitic chemical reactions, for instance those involving reactive oxygen.

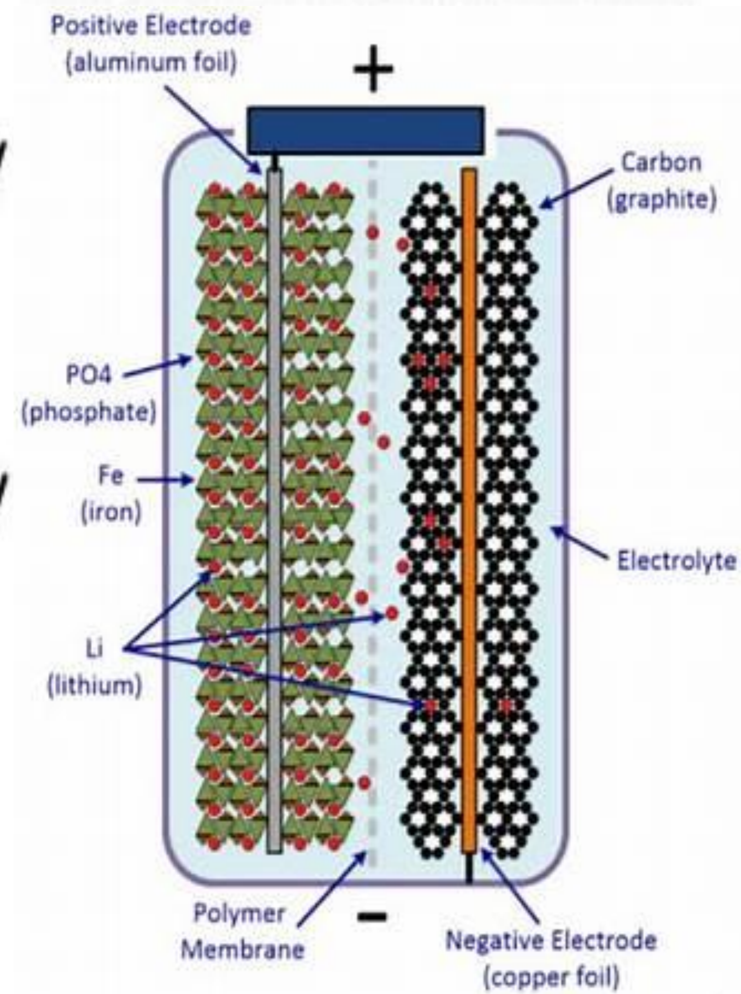
Al-air batteries

Lithium iron Phosphate battery

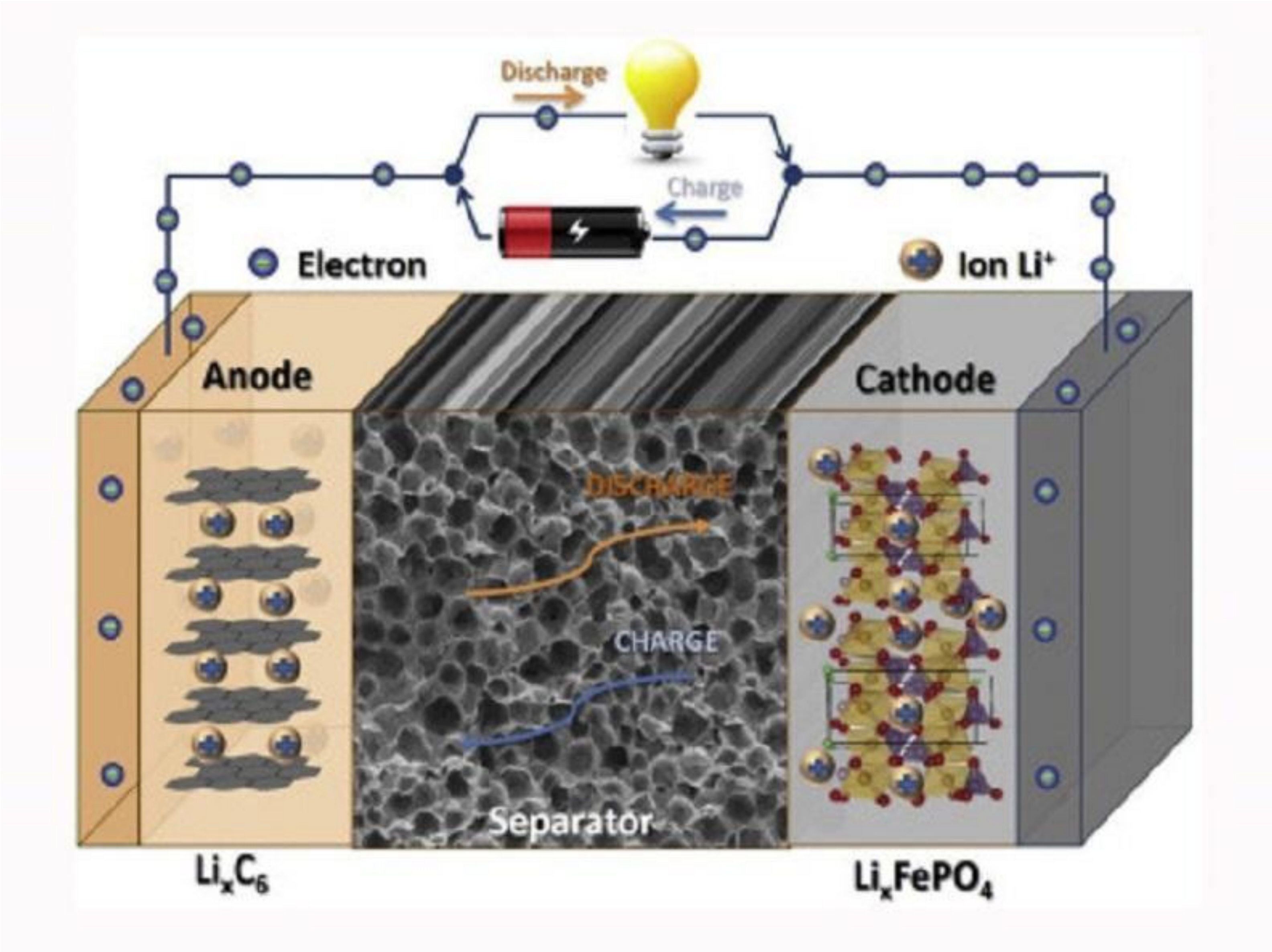
A lithium Ion rechargeable battery Use LiFePO4 as a anode material

A relatively new and revolutionary battery for Industrial and consumer electronics



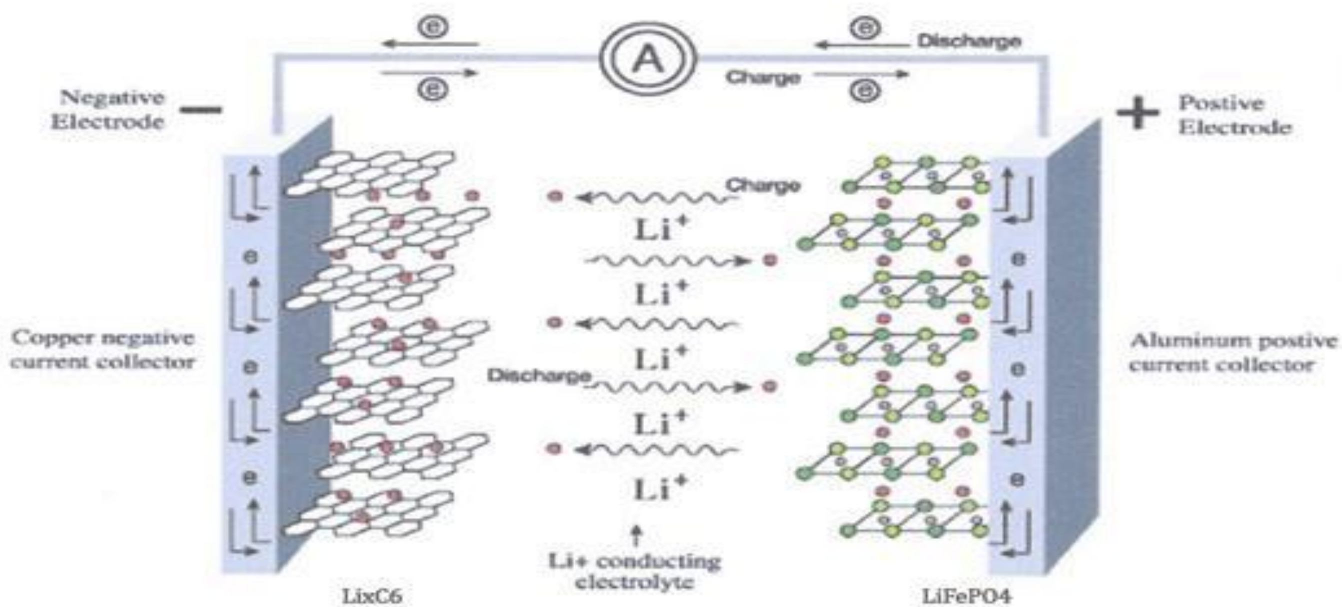


Lithium iron Phosphate battery



Lithium iron Phosphate battery

Chemical reaction equation of lithium iron phosphate battery pack



Positive Reaction: $LiFePO_4 \rightarrow Li_{1-x}FePO_4 + xLi + xe^-$

Negative Reaction: $6C + xLi^+ + xe^- \rightarrow Li_xC_6$

Total Reaction: $LiFePO_4 + 6xC \rightarrow Li_{1-x}FePO_4 + Li_xC_6$

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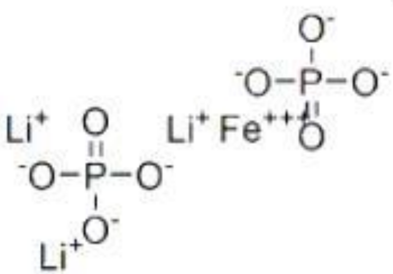
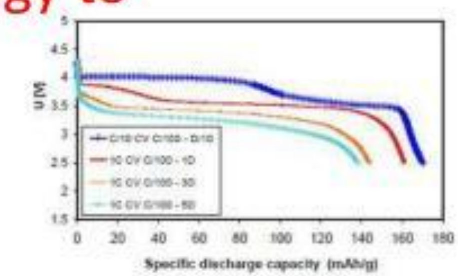
Features

LFP

* *Best Safety Characteristics*

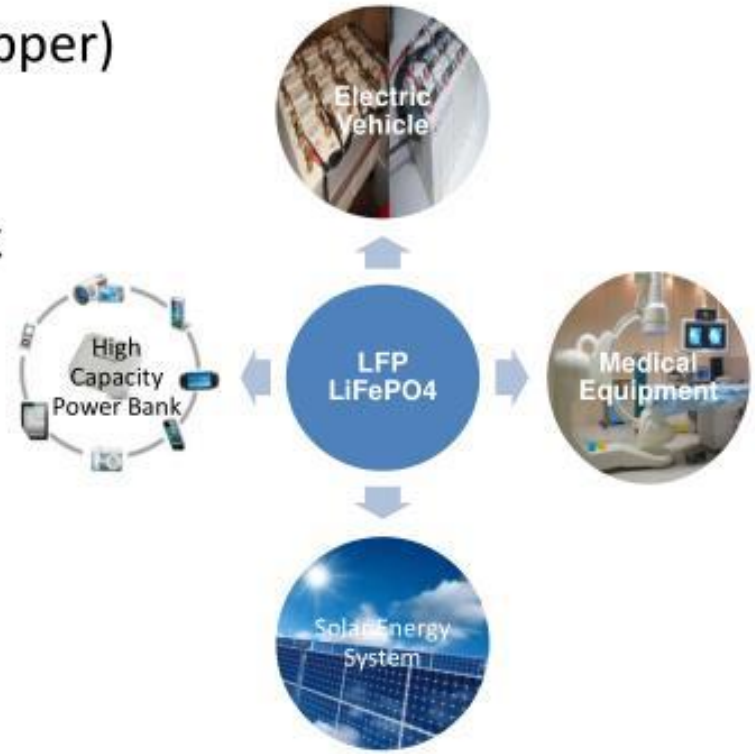
  (not required BMS, battery management system and protection module)
* Long cycle life (>2000 times)
* Good availability on material
* Higher discharge current
* Lower energy density (overcome by our proprietary Nano Technology to increase energy density)



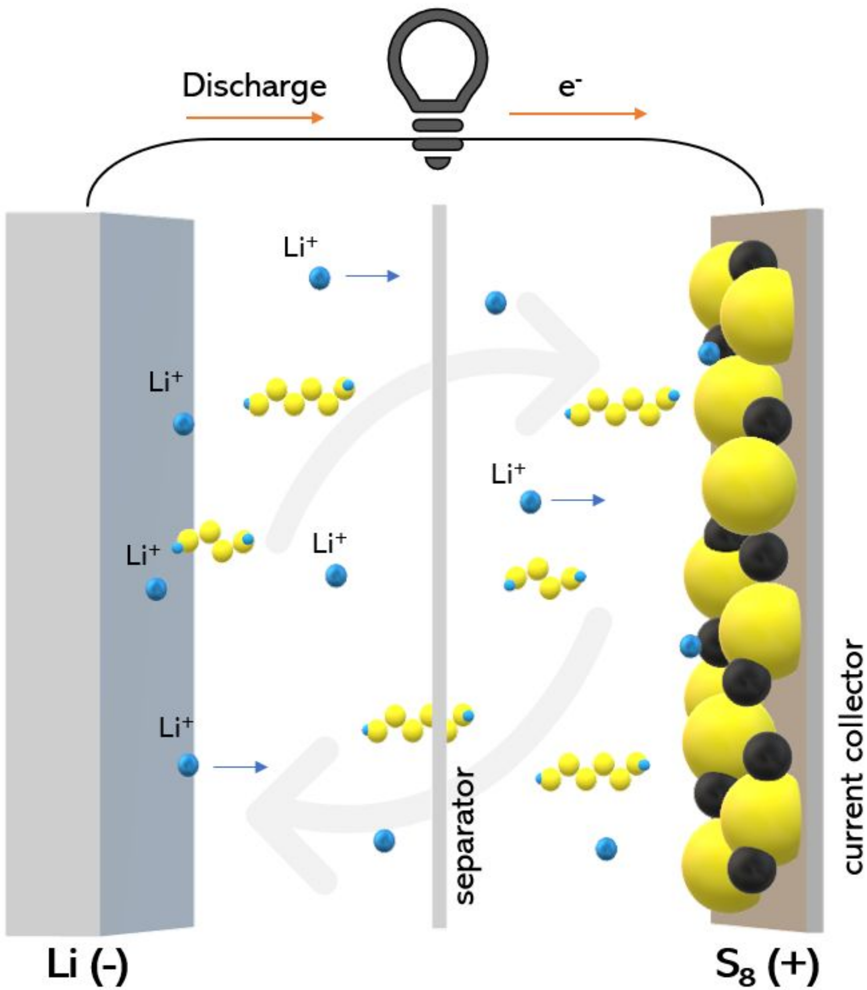


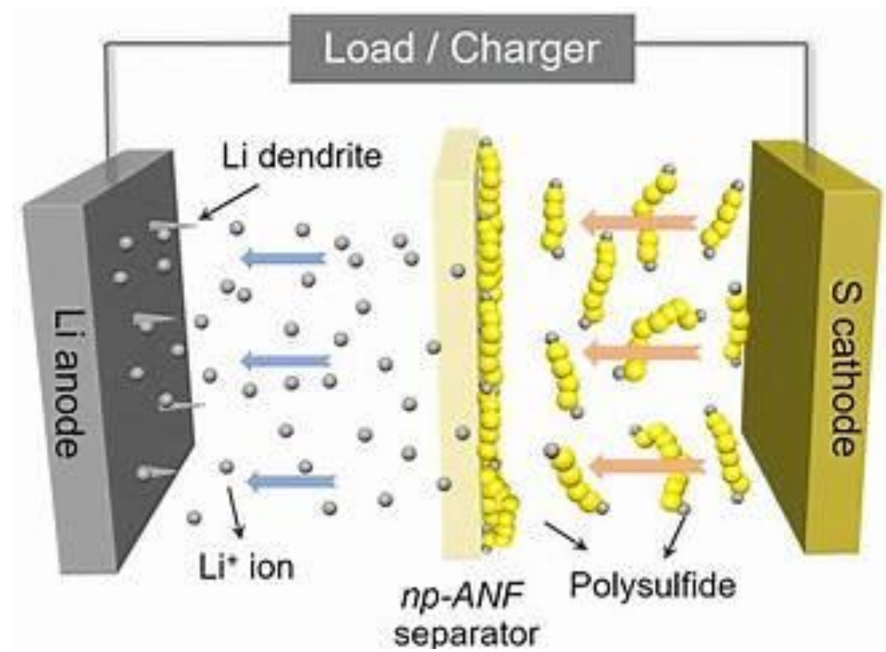
Lithium iron Phosphate battery : Applications

- Electric Vehicle (e-bike, e-scooter, e-car)
- Power tool(Drill, Saw, Cropper)
- UPS
- High Capacity Power Bank
- Emergency Lighting
- Solar Energy System
- Medical equipment
- etc

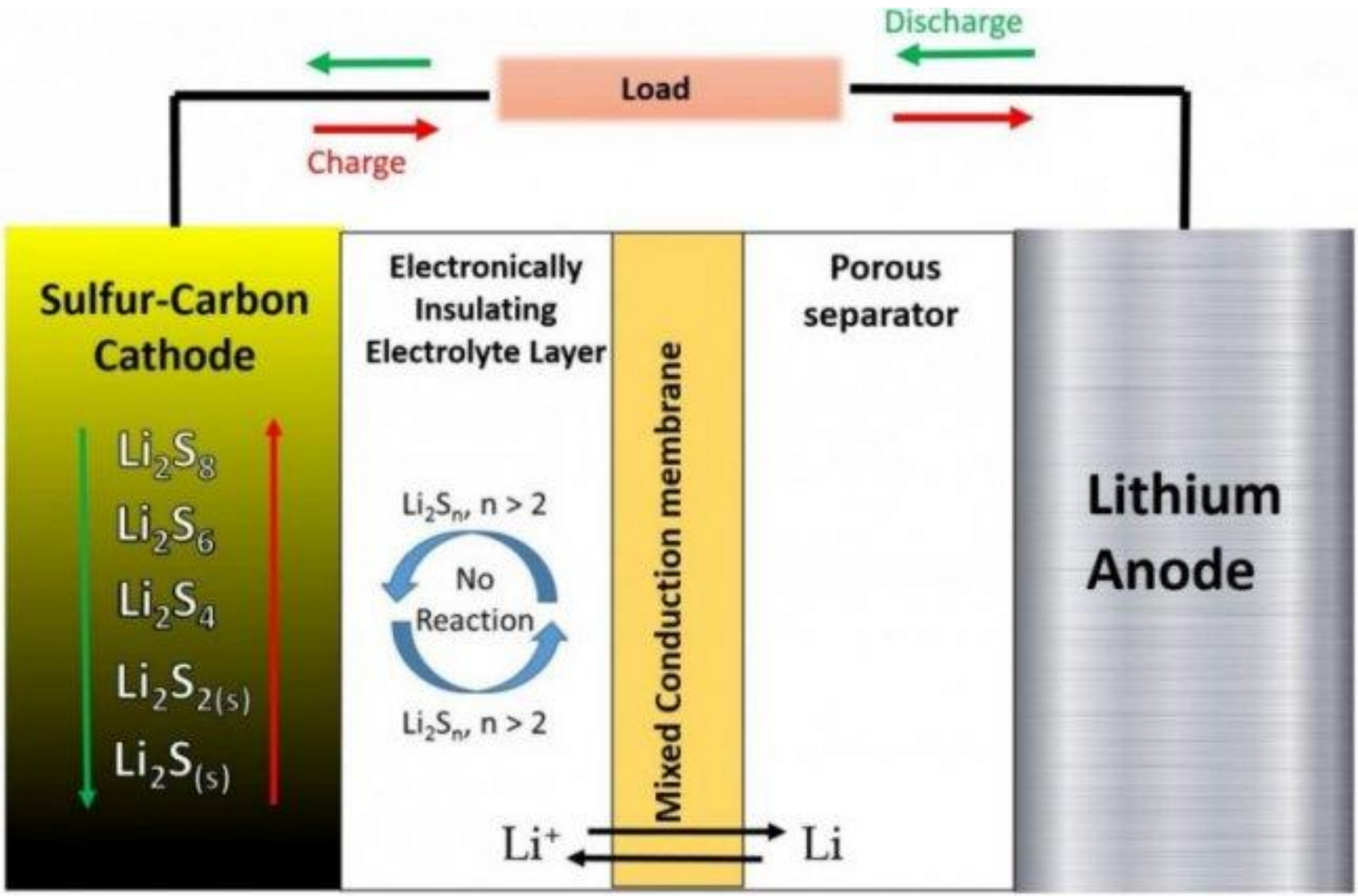


Lithium sulfur battery





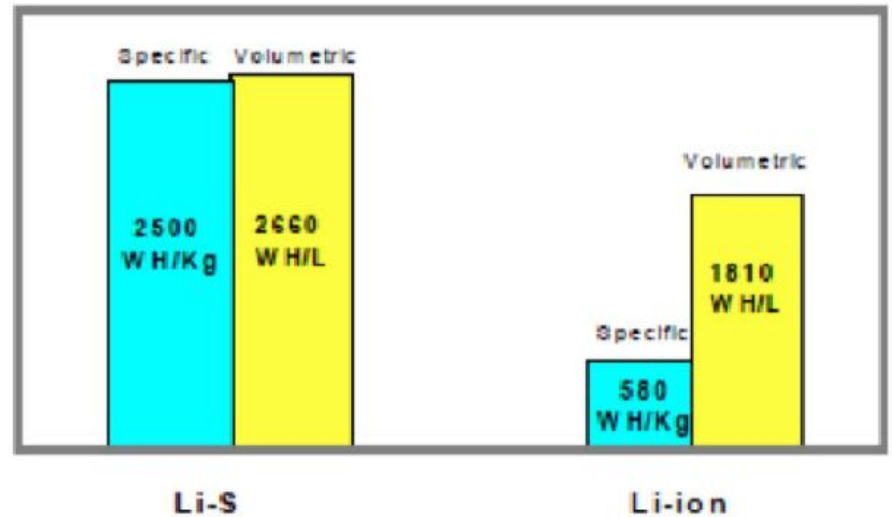
Lithium sulfur battery



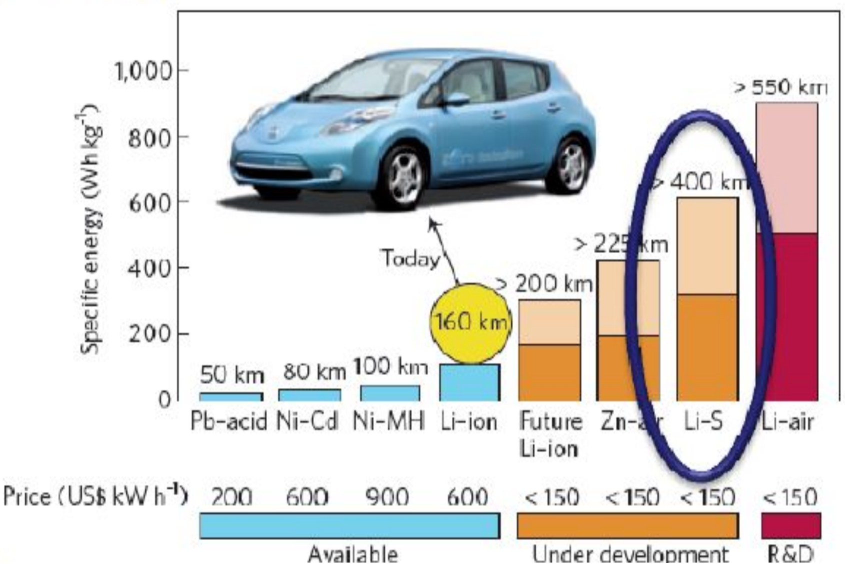
Why Lithium sulfur battery?

Potential impact of the innovation: *Li-S batteries are of great interest because:*

- **High energy density (3 times compared to Li-ion batteries)**
- **High capacity**
- **High rate capability**
- **Low cost**
- **Safety**



Theoretical energy density comparison
(Tudron F.B. *et al.* Sion Power Corporation, 2004).



Specific energies for rechargeable batteries

Lithium sulfur battery

- anode:

  Lithium

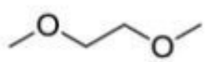
- cathode:

  Sulfur in carbon matrix

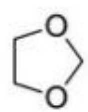
  (optional with polymeric binder)

- electrolyte

  DME-DOL-Mixture



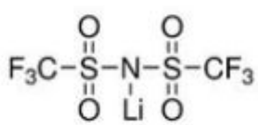
DME

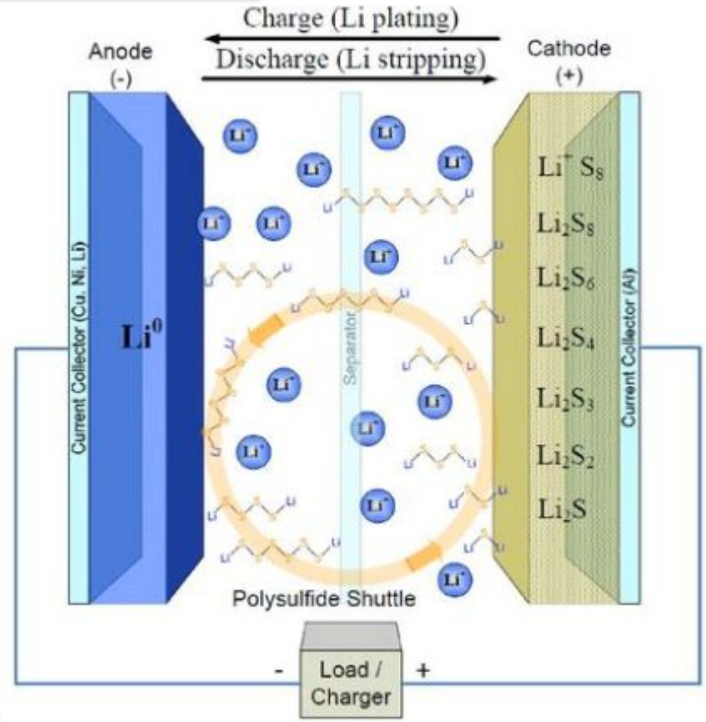


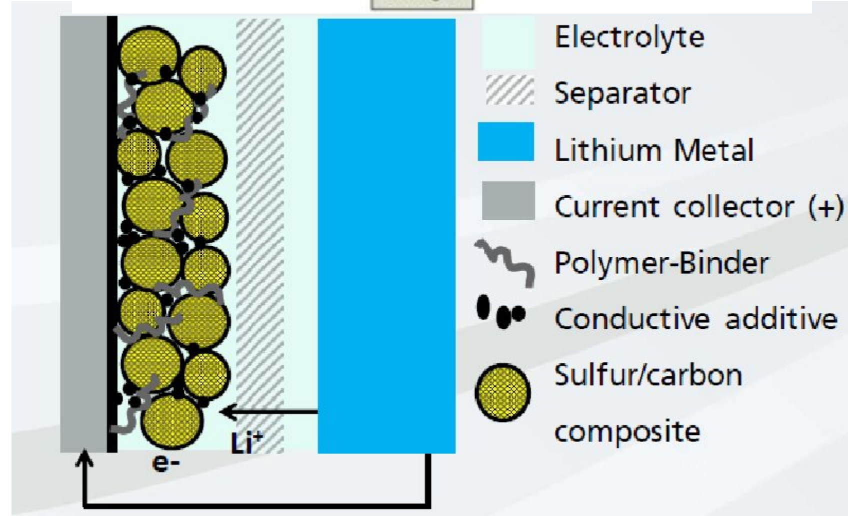
DOL

- conducting salt

  $LiN(SO_2CF_3)_2$ (LiTFSI)



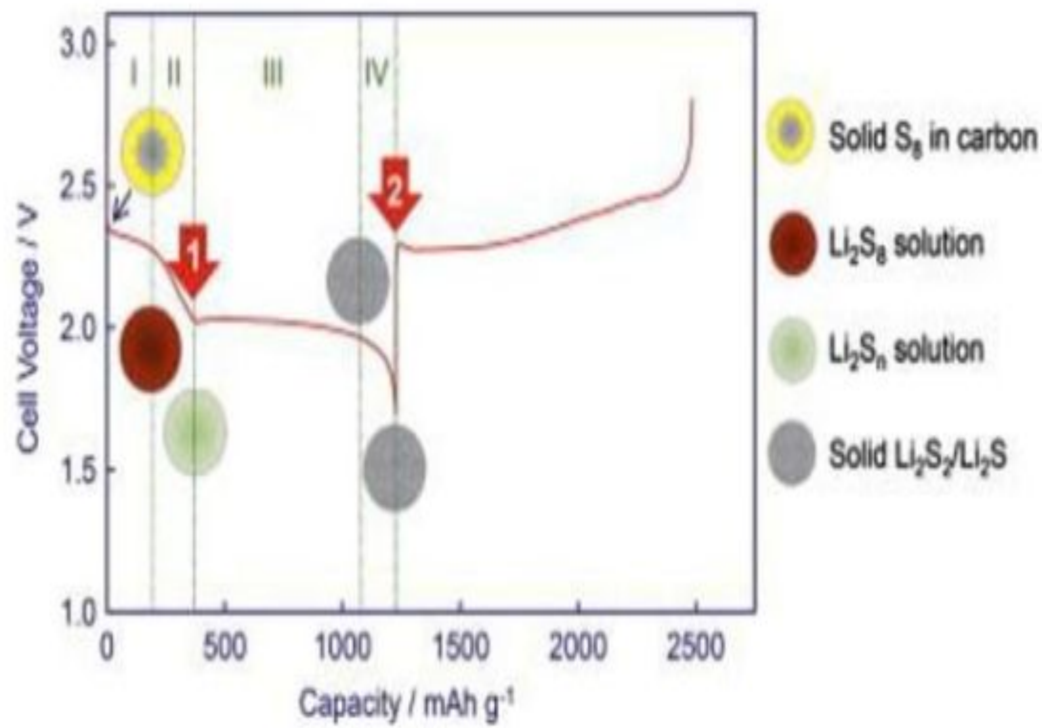





Fraunhofer
IWS
Schematical principle of a
lithium-sulfur battery

Lithium sulfur battery: Reactions

- anode
  - $Li \rightarrow Li^+ + e^-$
- cathode
  - $S_8 + 16\ e^- \rightarrow 8\ S^{2-}$
- redox equotation
  - $16\ Li + S_8 \rightarrow 8\ Li_2S$

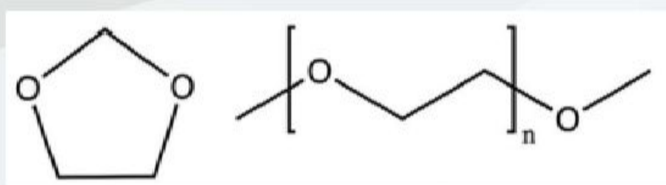


Electrolytes

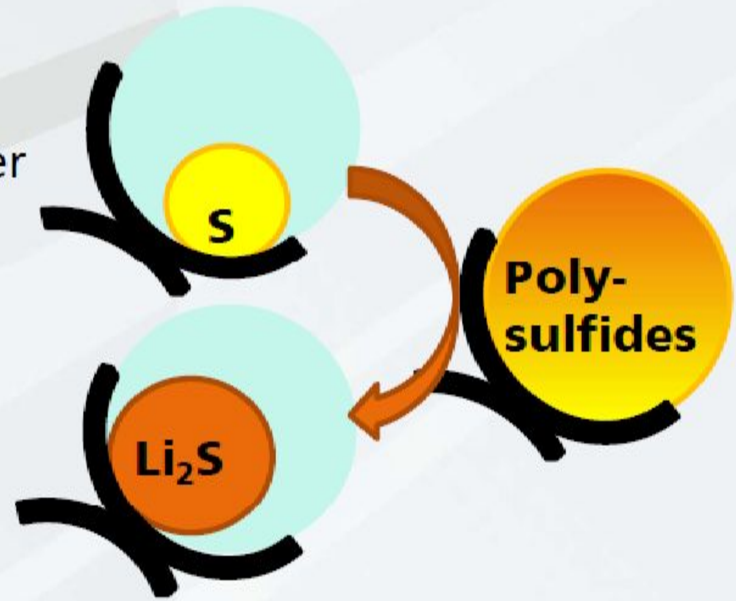
- Cathode mechanism in ether electrolyte requires dissolution of polysulfides
- Most studies use >> 5 µl Electrolyte / mg S

Saturated PS-solution in DME/DOL

$LiNO_3$: SEI former
LiTFSI: salt



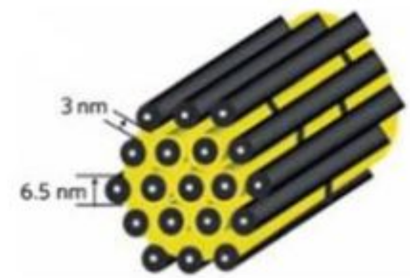
DOL Gn (n ≥ 1); DME (n = 1)



Lithium sulfur battery: Challenges

Problems

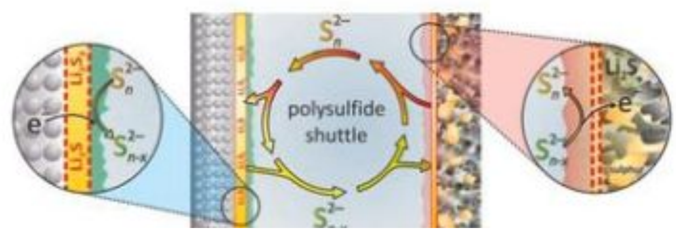
- poor electric conductivity of S and Li_2S



- expansion of volume during discharge process

- toxicity of conducting salt

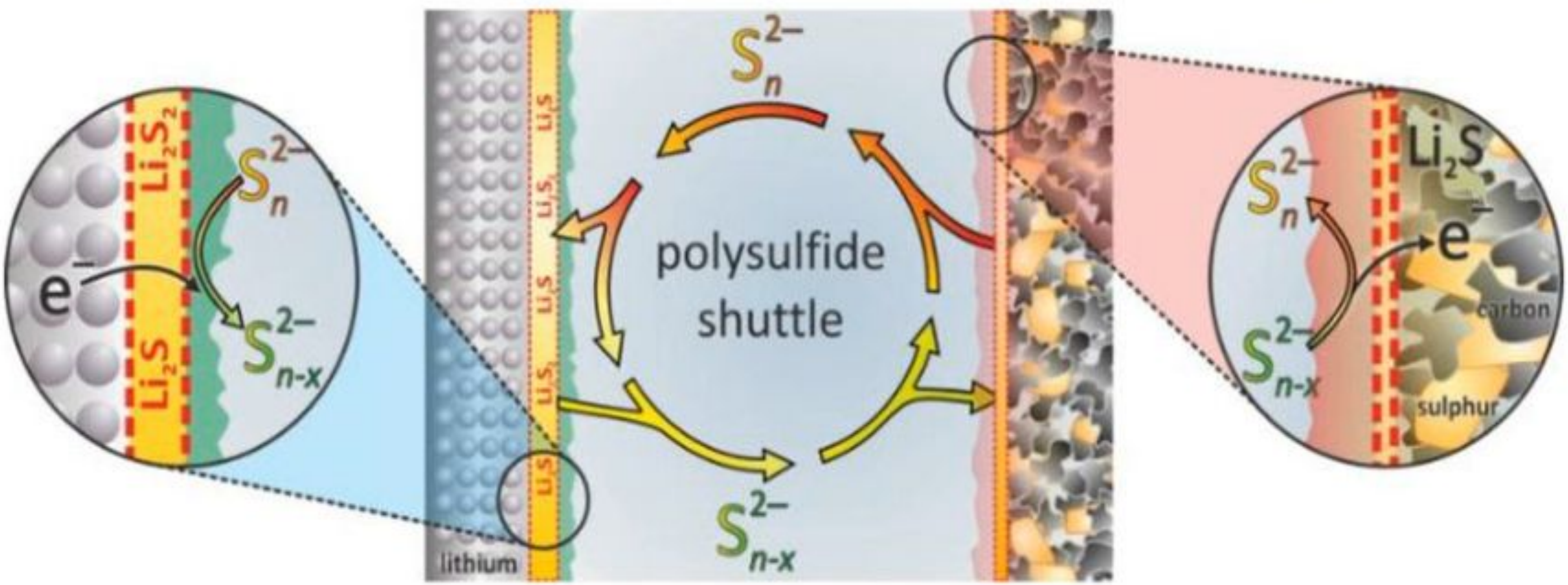
- polysulfide shuttle mechanism



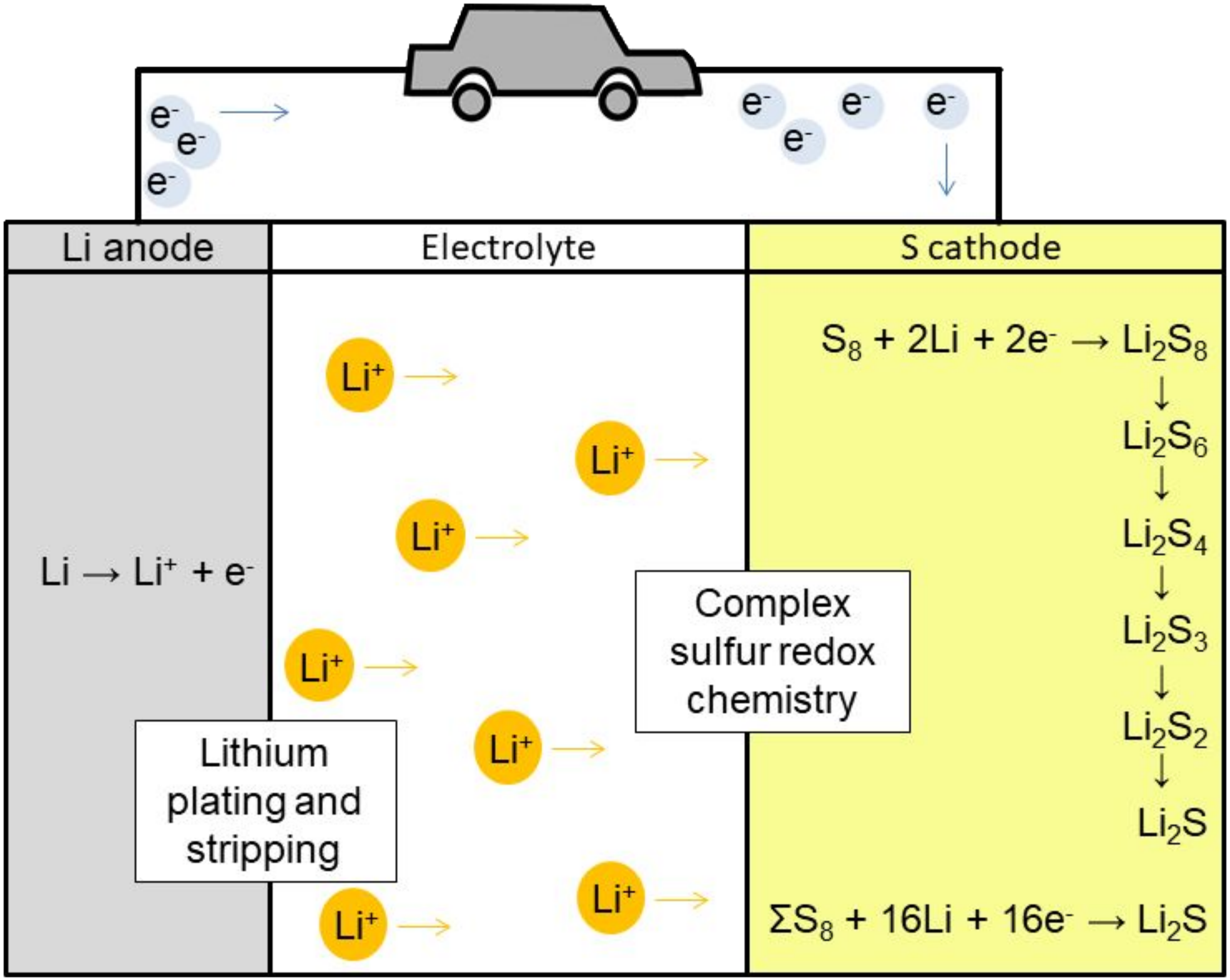
Lithium sulfur battery: Challenges

Polysulfide shuttle

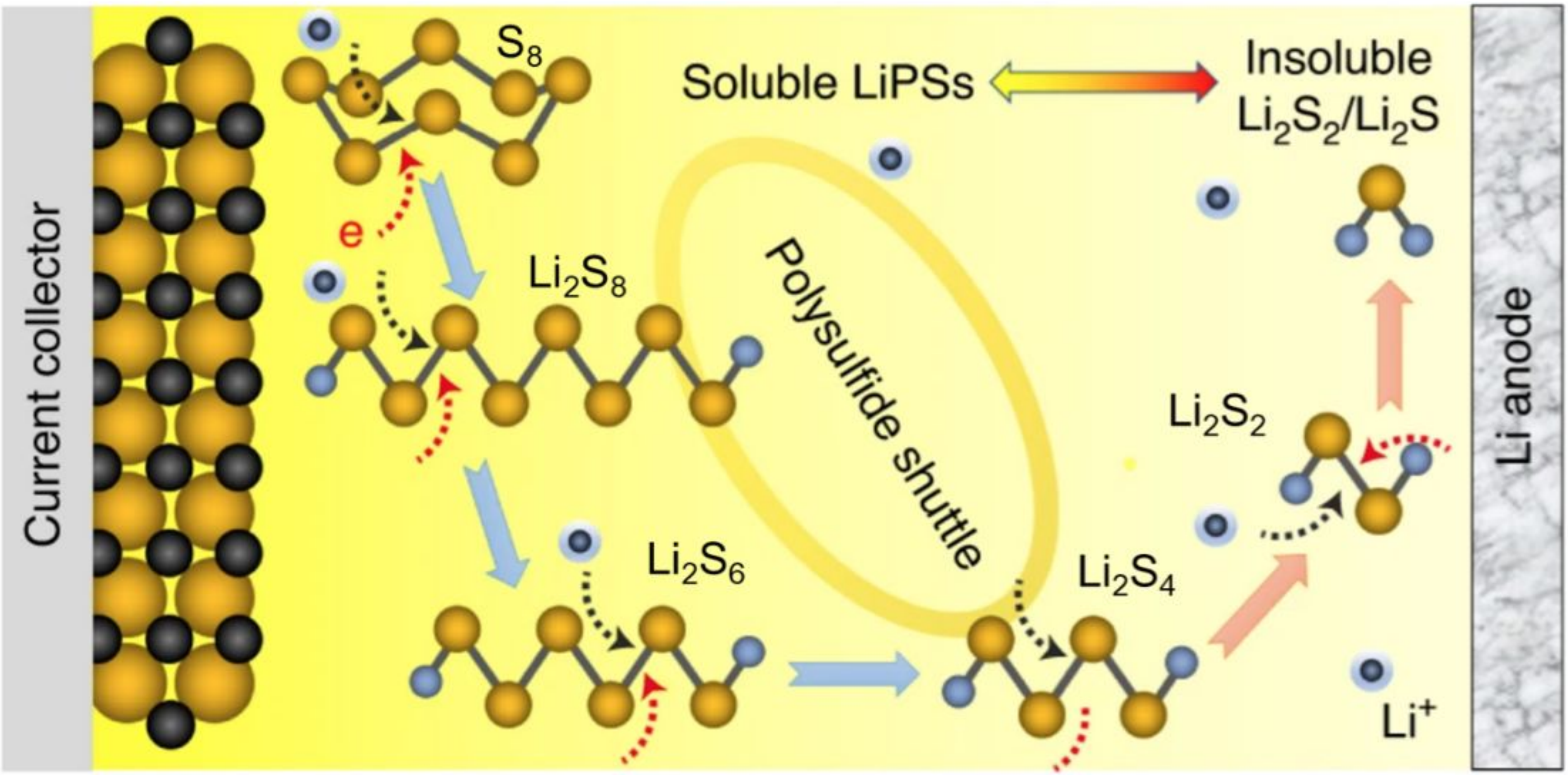
- loss of active mass
- morphological change of the cathode
- passivation of Li-Anode
- self-drain of battery
- increase of charge time



Lithium sulfur battery: Challenges



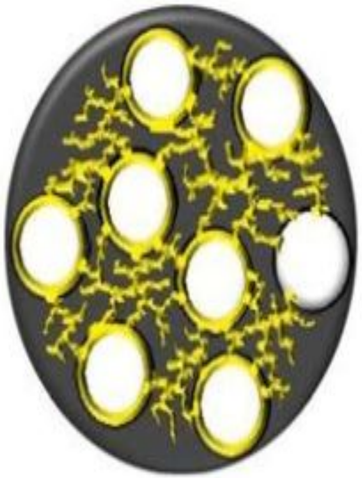
Lithium sulfur battery: polysulfide shuttle



Lithium sulfur battery: Challenges

Inhibition of shuttle mechanism

- addition of metaloxids
- various carbon nanostructures
- addition of $LiNO_3$
- solid electrolyte materials
- optimization of conducting salt

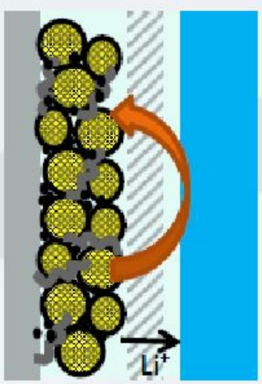


Lithium sulfur battery: Challenges

Main cause of low cycle life:
Lithium anode surface reactions

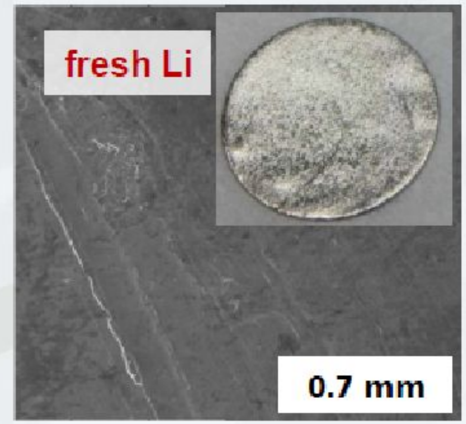
Degradation mechanisms

- Depletion of org. electrolyte, cells dry out
- Dendritic or porous structure causes shortcuts
- Self-discharge and active material loss through polysulfide shuttle

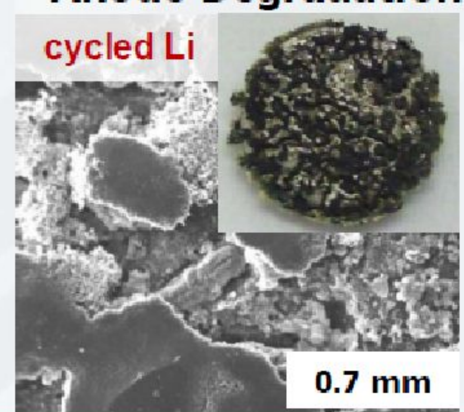


Polysulfide Shuttle:
Chemical reduction of dissolved PS at anode surface

→ $LiNO_3$-Additive suppresses shuttle

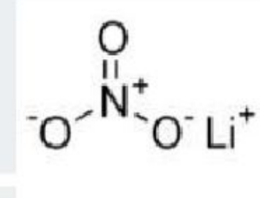


Anode Degradation



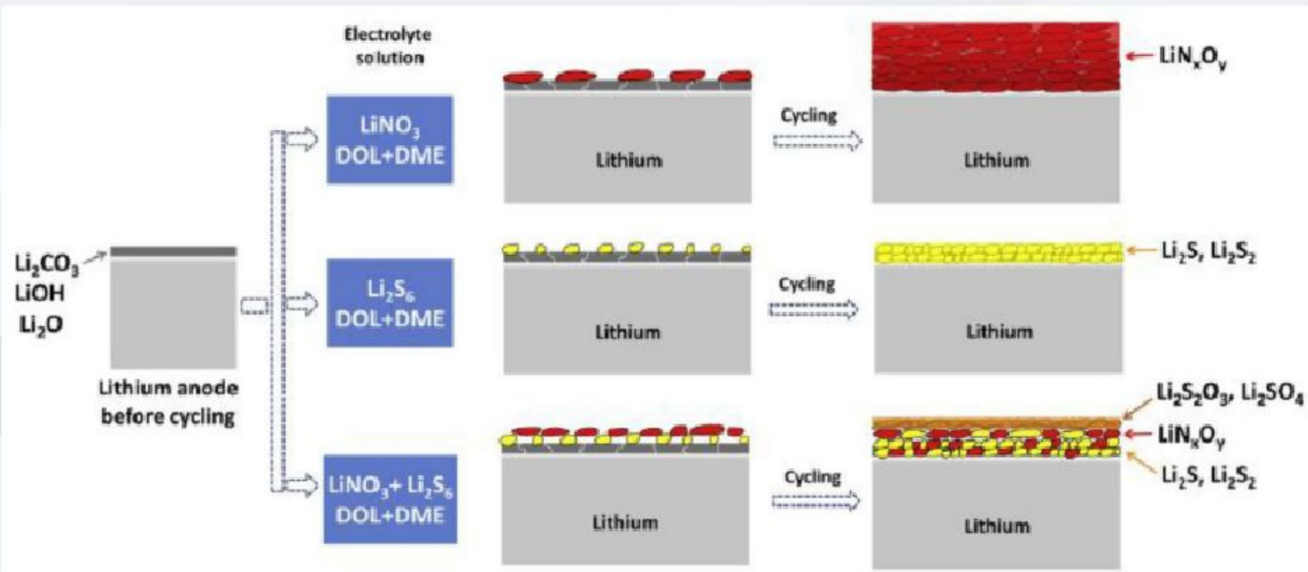
Electrolyte additives

The role of $LiNO_3$
Standard electrolyte additive in DME/DOL



$LiNO_3$: Directing SEI formation in Li-S cells

- Surface film: inorganic and organic species (LiN_xO_y, ROLi + $ROCO_2Li$)
- Compact and homogenous surface film formed with $LiNO_3$
- Enhanced stability of lithium anode and improved cycle life



LiS batteries pros and cons

Pros	Cons
Sulfur is abundant and inexpensive	Rapid capacity degradation after 300 cycles
Reduced risk of thermal runaway	Sulfur is an insulator, requiring additional conductive materials
Non-toxic and more sustainable than lithium-ion batteries	Polysulfide dissolution reduces efficiency

Lithium polymer battery

Significant criteria that distinguish Li-polymer batteries from other types of cell

Li-polymer cells, also known as soft or pouch cells, have a thin and somewhat ‘soft’ housing – like a pouch – made from deep-drawn aluminum foil.

The mostly prismatic housing can be produced more easily and cheaply than the hard cases of Li-ion cells. The other components, in wafer-thin layer foils (< 100 µm) can also be mass-produced at relatively low cost.

The cells are lightweight, thin and can be made in a wide range of shapes and sizes. Both large formats and heights of less than 1 mm can be achieved. The cells do, however, require careful mechanical handling.

Lithium polymer battery structures

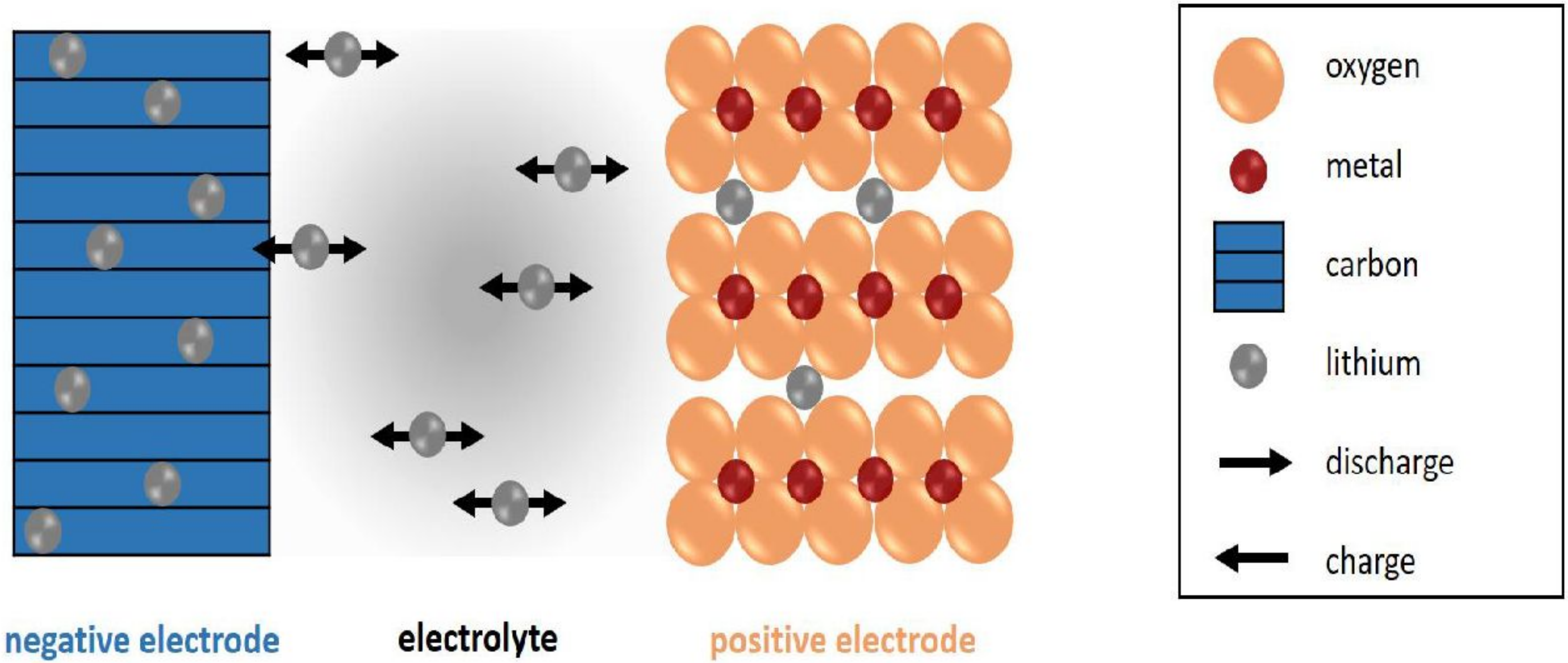
The lithium-polymer differentiates itself from conventional battery systems in the type of electrolyte used. The original design uses a dry solid polymer electrolyte. This electrolyte resembles a plastic-like film that does not conduct electricity but allows ions exchange

The dry polymer design offers simplifications with respect to fabrication, ruggedness, safety and thin-profile geometry. With a cell thickness measuring as little as one millimeter (0.039 inches), equipment designers are left to their own imagination in terms of form, shape and size.

Lithium polymer battery: Challenges

$$Li_xC_n \underset{L}{\overset{E}{\rightleftharpoons}} C_n + xLi^+ + xe^-$$

$$Li_{1-x}MO_2 + xe^- + xLi^+ \underset{L}{\overset{E}{\rightleftharpoons}} LiMO_2$$



$$\text{cell reaction:} \quad Li_{1-x}MO_2 + Li_xC_n \underset{L}{\overset{E}{\rightleftharpoons}} LiMO_2 + C_n$$

The manufacture of Li-polymer cells

- First, the electrode materials are mixed and prepared (material mixing).
- The mixture is pasted onto metallic substrates (coating).
- This is followed by drying, calendaring and slitting. Calendaring means rolling the material out to the desired thickness.
- The slitting process brings the electrode bands to their planned final dimension.
- Deflectors are attached to both electrodes (electrode assembly).
- They are wound around the core together with the separator, usually a three-layered polyolefin (winding). The core consists generally of a flat pin.
- The winding is placed in the cavity of a foil, which is partly folded and laid over the winding. The foil is welded to seal the sides.

The manufacture of Li-polymer cells

The packed winding must be dried in a vacuum under complete exclusion of moisture.

- Next is the injection of the electrolyte with the conductive salt, the degassing and the sealing of the cell.
- Since the cells are manufactured in a discharged state, they must now be formed and activated. The first charging takes place at this stage (electrical activation).
- The process is completed with tests of voltage over time (in which cells with short circuits are removed), capacity, quality and safety (x-ray full inspection and packaging inspection).

Advantages and limitations of Li-polymer cells

Advantages

- Very low profile - batteries resembling the profile of a credit card are feasible.
- Flexible form factor - manufacturers are not bound by standard cell formats. With high volume, any reasonable size can be produced economically.
- Lightweight - gelled electrolytes enable simplified packaging by eliminating the metal shell.
- Improved safety - more resistant to overcharge; less chance for electrolyte leakage.

Limitations

- Lower energy density and decreased cycle count compared to lithium-ion.
- Expensive to manufacture.
- No standard sizes. Most cells are produced for high volume consumer markets.
- Higher cost-to-energy ratio than lithium-ion